# ME1103
# Principles of Mechanics and Materials

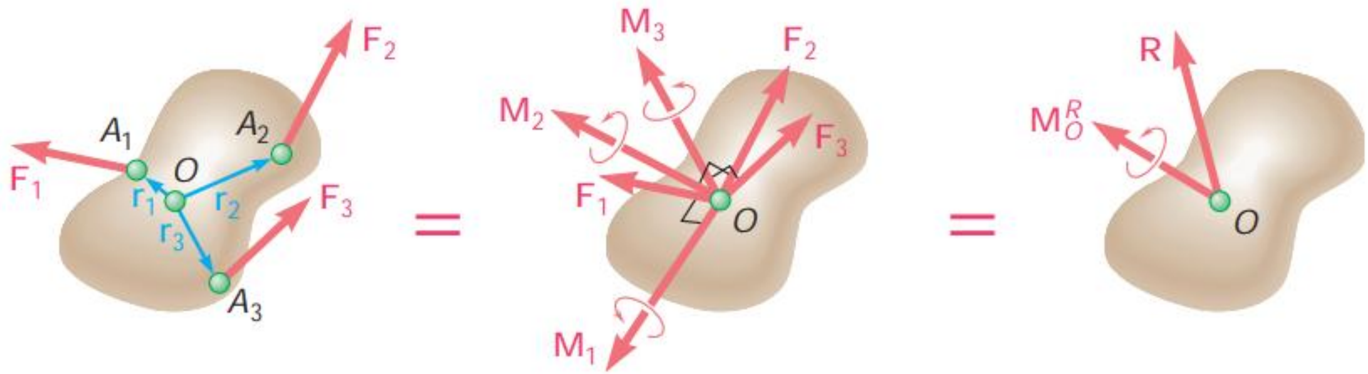
STATIC OF RIGID BODY

# Learning Outcomes

- Understand the concept of *static equilibrium* on *rigid body*.
- Able to differentiate *internal* vs *external* forces.
- Able to draw *free-body diagrams* and hence derive force and moment equations to solve problems in static equilibrium.
- Recognize the *different types of supporting conditions* and impose them appropriately to represent as reactions.

# Static Equilibrium of Rigid Body

- Recall that the effect of a system of forces and moments can always be reduced to an equivalent resultant *force-couple pair* at any given point.




- For the *rigid body* to be in *static equilibrium,* the resultant force and moment must be equal to zeros for any reference point, i.e.

$$\vec{R} = \sum \vec{F}_i = 0 \ \ and \ \ \vec{M}_O^R = \sum \vec{M}_i = 0$$

# Rigid Body with Concurrent Forces

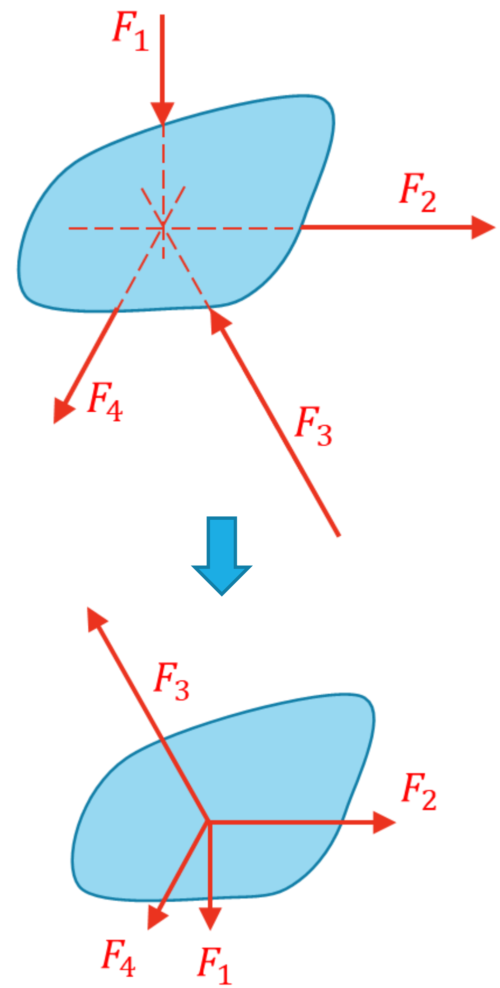
- Suppose the applied forces acting on the body are ***concurrent***, i.e. their line-of-actions coincide at a common point.

- In this case, the body will attain its static equilibrium as long as the force equations are satisfied, i.e.

$$R_x = R_y = R_z = 0$$

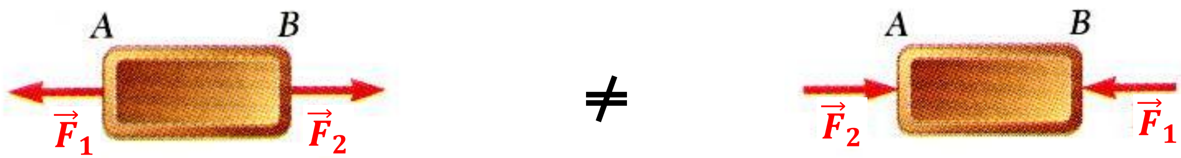
  As such, the problem can be regarded as ***statics analysis of a particle***.

- Notice also that the forces can be shifted along their line of actions, say to the common point *D*, to form another equivalent system of forces. Such actions can be done because of the ***principle of transmissibility***.



# Principle of Transmissibility

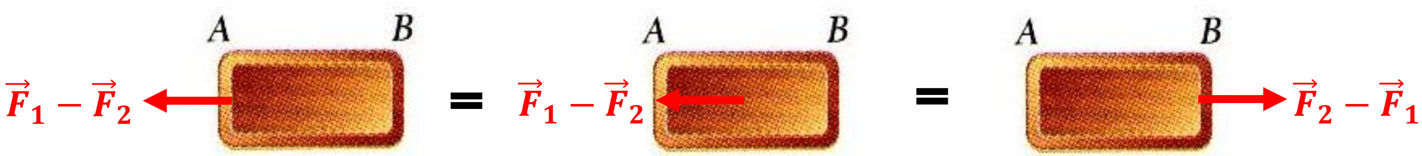
➢ Consider a block that is subjected to pulling forces and pushing forces as shown.




Tensile state

Compressive state

However, if the block is a ***Rigid Body***...

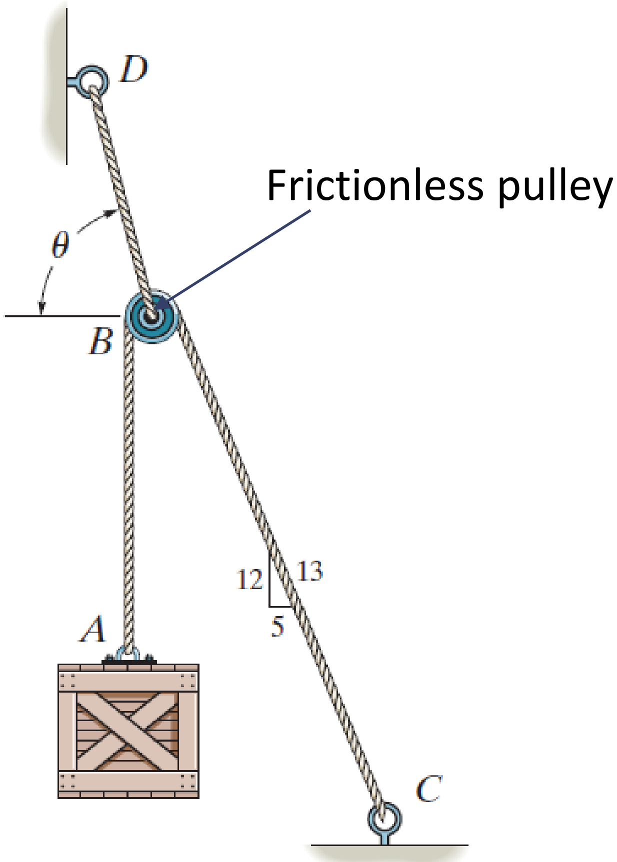



➢ *Principle of transmissibility* states that the conditions of equilibrium or motion of a *rigid body* will remain unchanged if the forces slide along their line of actions.

➢ This implies that the *point of actions of the forces are not important*, and hence we have the flexibility to *slide the force along its line-of-action* to another point for the convenience of calculations.

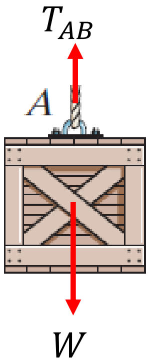
# Example 2.1

The cords *ABC* and *BD* can each support a maximum load of 100 *N*. Determine the maximum weight of the crate, and the angle $\theta$ for equilibrium.

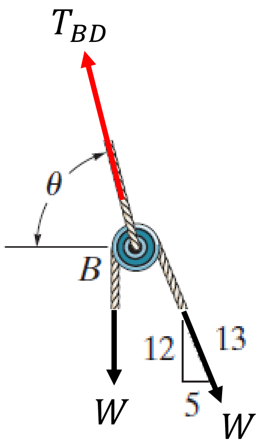


# Example 2.1…

Looking at the crate, it should be obvious that the tension in the vertical chord *AB* must balance the weight of the crate. That is,




$$T_{AB} = W$$




Next, consider the forces acting on pulley *B.*

Given that the pulley is frictionless, the same tension $T_{AB}$ runs throughout the entire rope *ABC*. Hence, the force equations are:

$$W\left(\frac{5}{13}\right) - T_{BD}cos\theta = 0 \Rightarrow T_{BD}cos\theta = \frac{5}{13}W$$

$$T_{BD}sin\theta - W - W\left(\frac{12}{13}\right) = 0 \Rightarrow T_{BD}sin\theta = \frac{25}{13}W$$

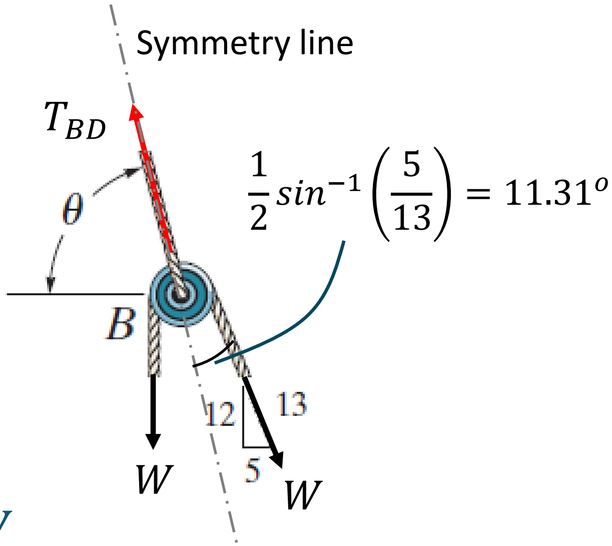
Dividing the two equations, gives

$$tan\theta = 5 \Rightarrow \theta = \mathbf{78.69^o}$$

# Example 2.1…

Alternatively, due to the symmetrical loadings, we can also determine the angle $\theta$ to be

$$\theta = 90^o - 11.31^o = \mathbf{78.69^o}$$




At this given angle, the tension in *BD* is

$$T_{BD} = \frac{25}{13sin(78.69^o)}W = 1.961W$$

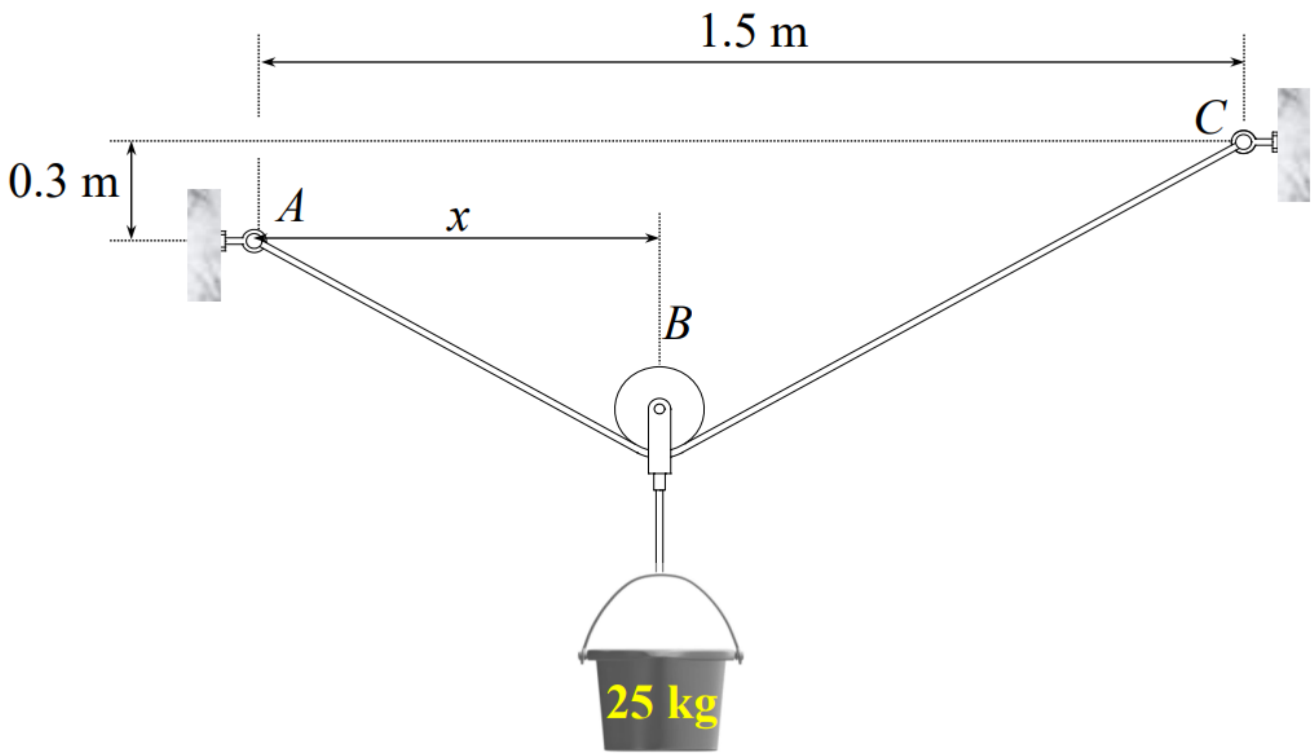
This suggests that the tension in cord *BD* will reach the maximum load of 100 *N* before that in cord *ABC,* which is only *W*.

With this observation, the maximum weight of the crate can be calculated as

$$W = \frac{100}{1.961} = \mathbf{51.0\ N}$$

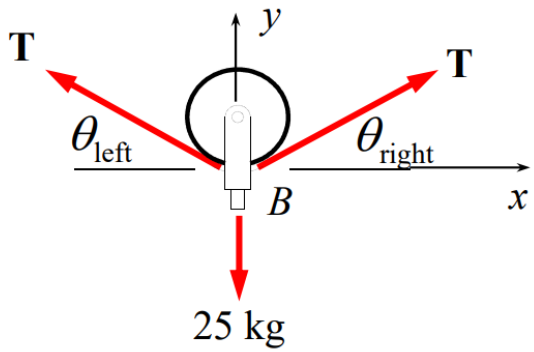
# Example 2.2

Given the total length of cable *ABC* is 2.5 meters, determine *x* and tension in the cable when the system is in static equilibrium. Neglect the mass of the and assume it is frictionless.



# Example 2.2…

Consider the FBD of the pulley




Force equilibrium in x-direction, gives

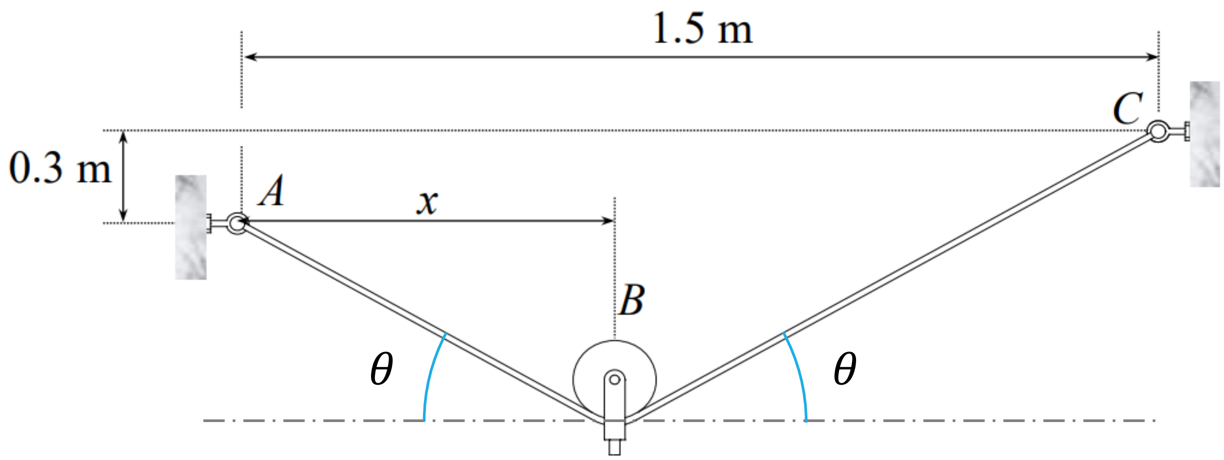
$$-Tcos(\theta_{left}) + Tcos(\theta_{right}) = 0$$

$$\Rightarrow \theta_{left} = \theta_{right} = \theta$$

Force equilibrium in *y*-direction, gives

$$2Tsin\theta - 25(9.81) = 0 \Rightarrow T = \frac{137.34}{sin\theta}$$

# Example 2.2…



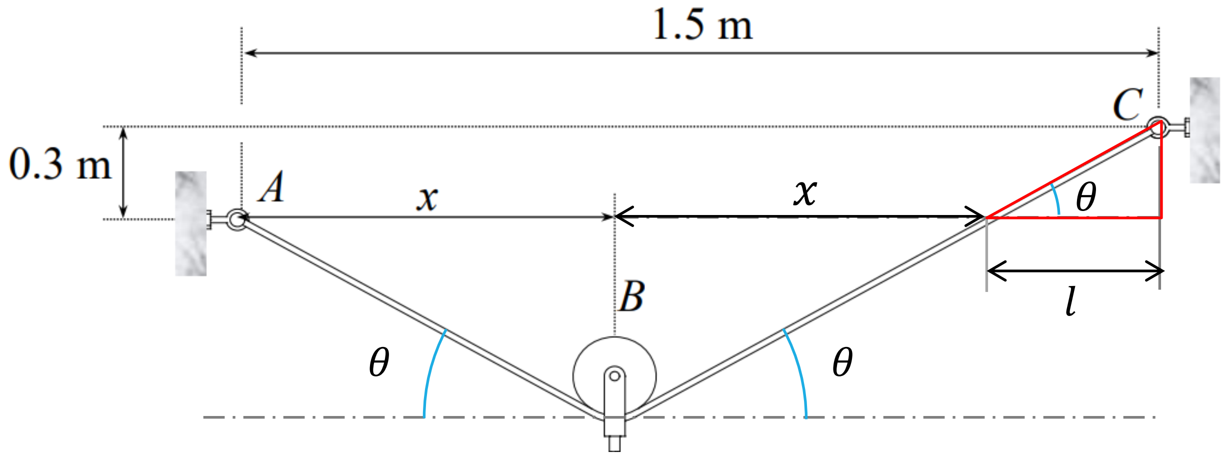

Now based on the geometrical configuration, we get

$$\frac{x}{cos\theta} + \frac{1.5 - x}{cos\theta} = 2.5 \Rightarrow cos\theta = \frac{1.5}{2.5} \Rightarrow \theta = 53.15^o$$

Substituting $\theta = 53.15^o$ into the tension expression, gives

$$T = \frac{137.34}{sin53.15^o} = \mathbf{153.3\ N}$$

# Example 2.2...



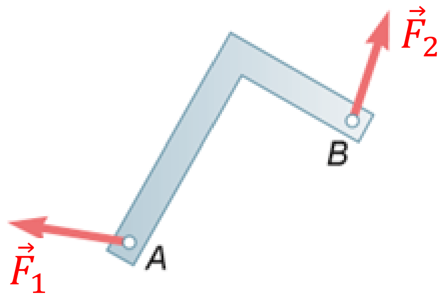

Based on the geometry of the small right-angled triangle,

$$tan\theta = \frac{0.3}{l} \Rightarrow \; l = \frac{0.3}{tan\theta} = 0.2248$$

With this observation,

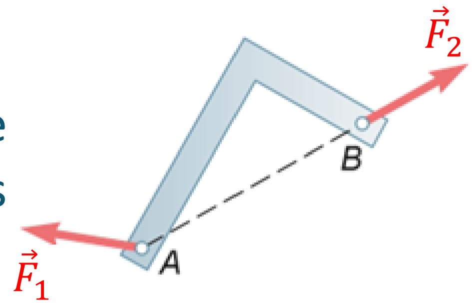
$$2x + 0.2248 = 1.5 \Rightarrow x = \mathbf{0.6376\,m}$$

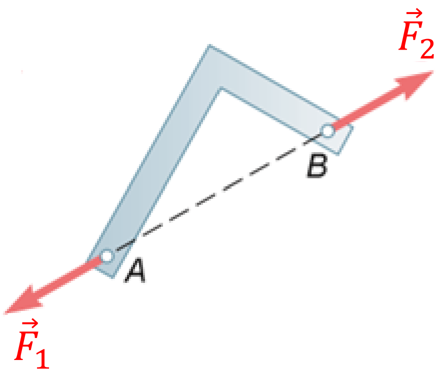
# Force in Statics - Two-force body




- Consider a rigid body subjected to only *two* forces $\vec{F}_1$ and $\vec{F}_2$.

- For static equilibrium, the moment about *A* must be zero. This means the line of action of $\vec{F}_2$ must pass through *A*.

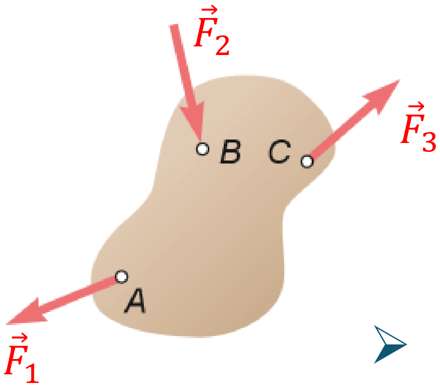



- For the same reason, i.e. the moment about *B* is also zero, suggests that the line of action of $\vec{F}_1$ must pass through *B.*

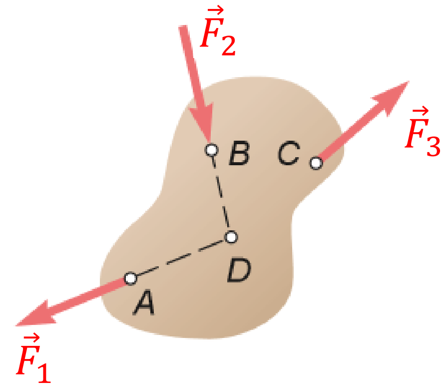
- Finally, the force equilibrium means that the two forces must be *equal in magnitude and opposite in direction, i.e.* $\vec{F}_1 = -\vec{F}_2$.

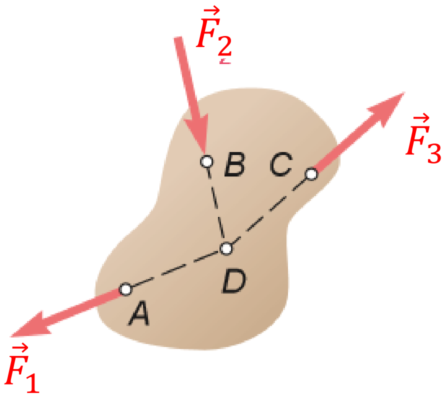
# Force in Statics - Three-force body




- Consider a rigid body subjected to only <u>*three*</u> forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ that are <u>*not parallel*</u>.

- The line of actions of $\vec{F}_1$ and $\vec{F}_2$ intersect at point *D*, at which the two forces will have zero moments.



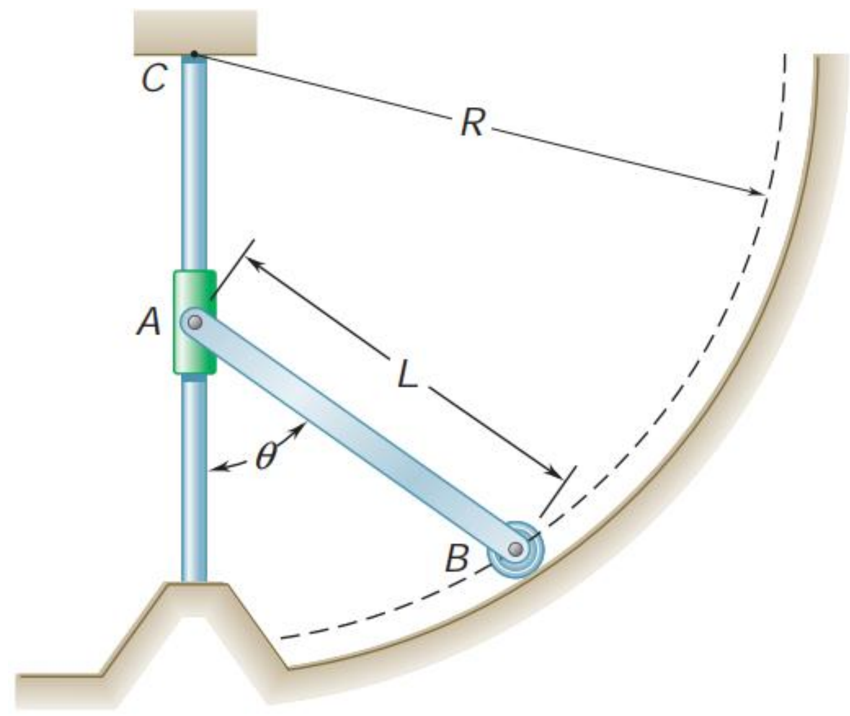

- To attain static equilibrium, $\vec{F}_3$ must also have zero moment about *D*. This means that its line of action must also pass through *D*.

- In summary, for a rigid body subjected to three forces to be in static equilibrium, the forces must be <u>*concurrent*</u>, and they formed a <u>*closed force-triangle*</u>.

## Example 2.3

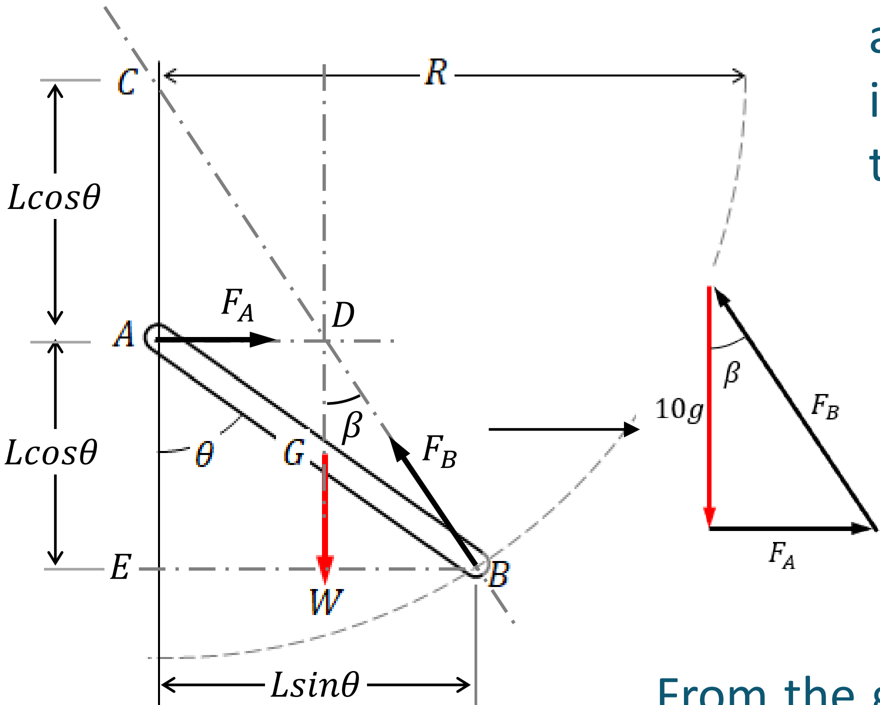
A slender rod of length $L$ and weight $W$ is attached to a collar at $A$ and is fitted with a small wheel at $B$. Knowing that the wheel rolls freely along a cylindrical surface of radius $R$ and neglecting all friction.

Given that $L$ = 1.5 $m$, $R$ = 2 $m$, and the rod is 10 $kg$, determine the reactions at $A$ and $B$.



# Example 2.3…

Consider the *FBD* of the rod.



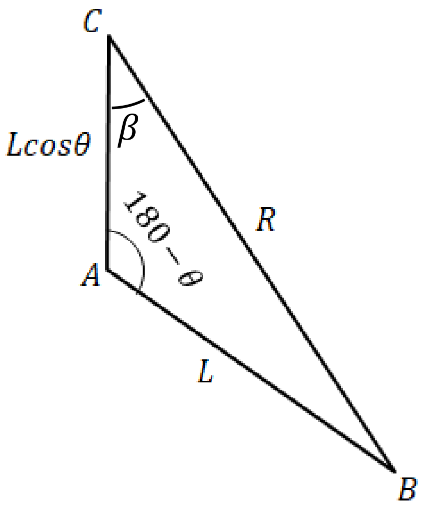

Since it is assumed that all the contacts among the components are frictionless, this implies that contact forces are perpendicular to the contact surfaces.

Hence, the forces at *A* and *B* are drawn as shown. Since this is a *three-force body*, their lines of actions must coincide at point *D*.

From the geometry, it can be observed that $\Delta ACD$ and $\Delta ECB$ are similar triangles, with $AC = AE = Lcos\theta$.

And lengths of *AB* and *CB* are *L* and *R*, respectively.

# Example 2.3…




Finally, using cosine rule to $\Delta ACB$, we get

$$R^2 = L^2 + (Lcos\theta)^2 - 2L(Lcos\theta)cos(180 - \theta)$$

$$\Rightarrow 3L^2cos^2\theta = R^2 - L^2$$

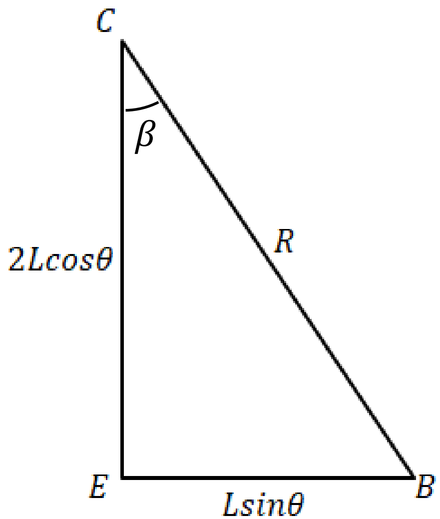
$$\Rightarrow cos^2\theta = \frac{R^2 - L^2}{3L^2}$$

Alternatively, apply Pythagoras theorem to $\Delta ECB$, i.e.

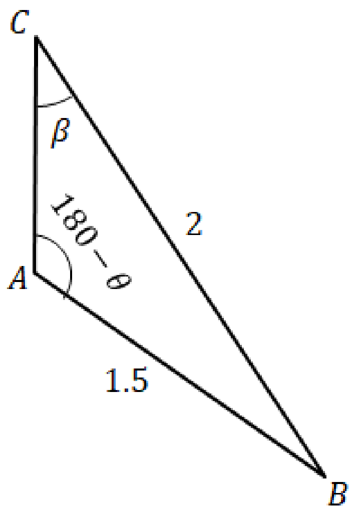
$$(Lsin\theta)^2 + (2Lcos\theta)^2 = R^2$$

$$\Rightarrow L^2(1 - cos^2\theta) + 4L^2cos^2\theta = R^2$$

$$\Rightarrow cos^2\theta = \frac{R^2 - L^2}{3L^2}$$



# Example 2.3…




Substitute the numeric for *L* and *R* into the expression, gives

$$\Rightarrow cos^2\theta = \frac{2^2 - 1.5^2}{3(1.5)^2} \Rightarrow \theta = 59.39^o$$

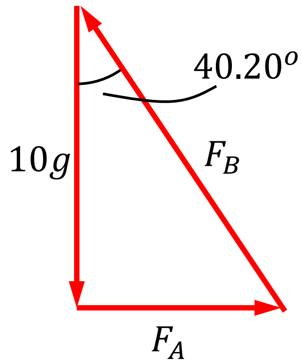
And applying sine rule to $\Delta ACB$, gives

$$\frac{sin\beta}{1.5} = \frac{sin(180 - 59.39)}{2} \Rightarrow \beta = 40.20^o$$

With the angular positions, the force triangle can be derived, which gives

$$\frac{F_A}{10g} = tan(40.20^o) \Rightarrow F_A = \mathbf{8.45g}$$

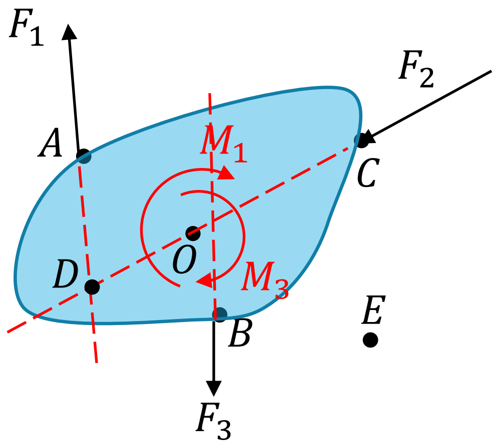
$$\frac{F_A}{F_B} = sin(40.20^o) \Rightarrow F_B = \mathbf{13.09g}$$



# Static Equilibrium of Rigid Body - 2D problem…

- Recall that if the forces acting on a body are *concurrent forces*, and they sum up to zero, then the body will be in *static equilibrium*.

$$R_i = \sum \vec{F}_i = 0, \qquad for\ i = x, y$$




- However, if the forces are not concurrent, they may give rise to *moments* that can cause the body to *rotate*.

- Hence, for the rigid body to reach its *static equilibrium*, the *resultant moment about any point of reference* must also be equal to zero, i.e.

$$\boxed{\sum M_A = \sum M_B = \sum M_C = \sum M_D = \cdots \sum M_O = 0}$$

# Static Equilibrium of Rigid Body – 2D problem…

- For 2D problems in rectangular coordinates, we generally impose the force and moment equilibrium conditions as follows:

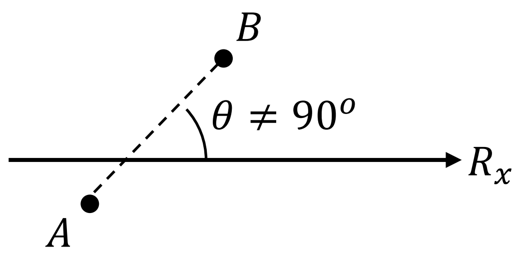
$$\boxed{R_x = R_y = M_A^R = 0}$$

where the moment can be taken about at any point in the 2D plane.

- We can also write the following three equations

$$R_x = 0 \;\; or \;\; R_y = 0, \qquad and \quad M_A^R = M_B^R = 0$$

which comprises *one force equation* and *two moment equations.*



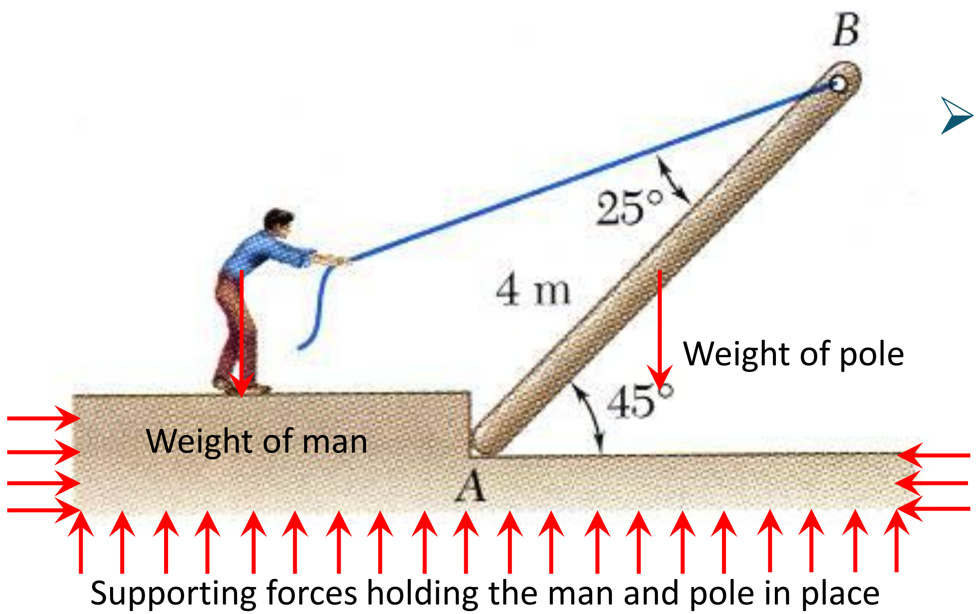

- Or we can also write *three moment equations*, i.e.

$$M_A^R = M_B^R = M_C^R = 0, but\ A, B\ anc\ C\ cannot\ be\ collinear.$$

**There can only be *THREE Independent Equations* from static analysis for *2D problems*.**

# Free Body Diagram

- To derive the force and moment equilibrium equations of a rigid body, it is often useful to create the ***free body diagram*** (*FDB*). It is a graphical representation to visualize the ***external applied*** forces, moments and resultant reactions on a body.



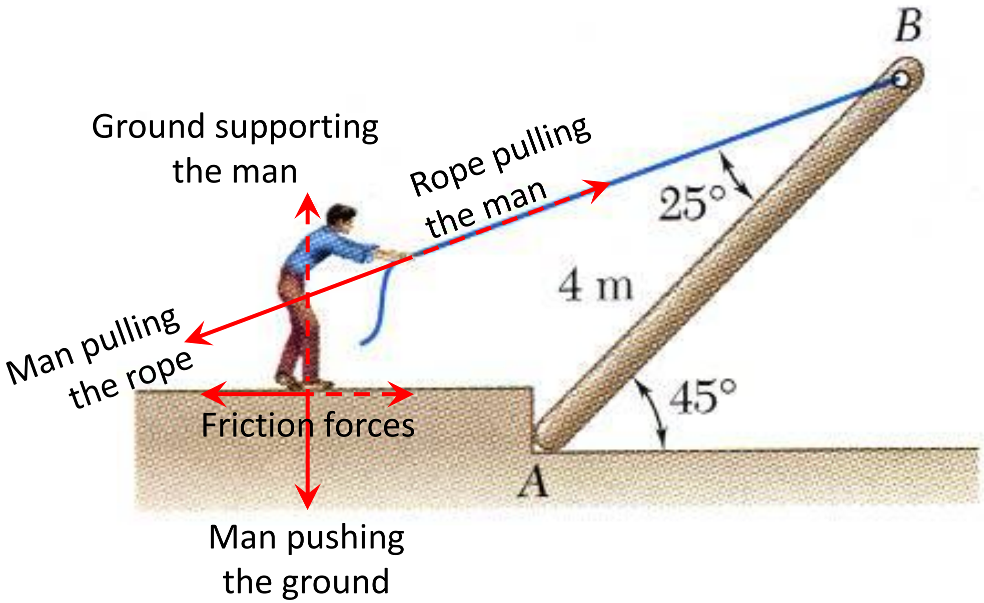

- Consider this example, the weights of the man and pole are forces due to gravity pull. They are ***external forces***.

- In addition, there must also be ***external*** supporting forces on the ground to hold the system in place.

- ***External forces*** - the action of other bodies on the body under consideration, which are responsible for the ***external behaviour*** of the body/system. For static equilibrium, the effective force and moment due to these forces are zeros.

# Free Body Diagram...

- On the other hand, we also know that the man is pulling one end of the rope, while the other end is supporting the pole. They give raise to the tension in the rope, which is an *internal force*.

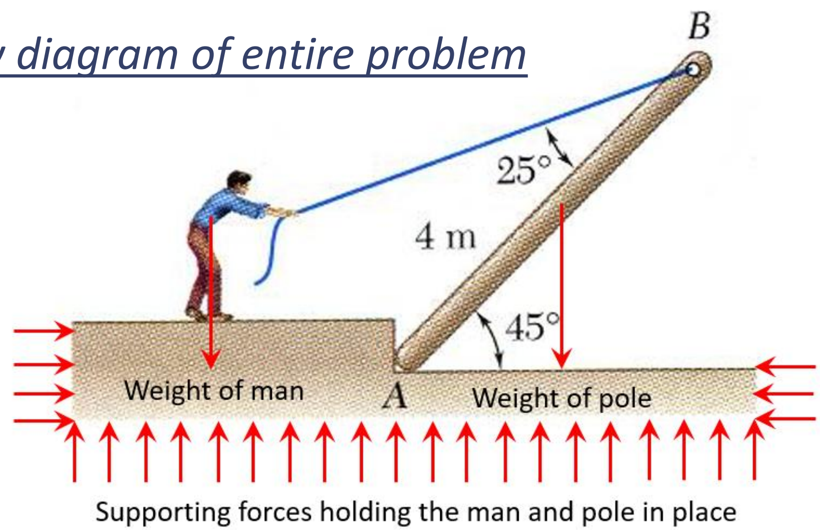



- While the man is pulling the rope, it can also be said that the rope is pulling the man, but in the opposite direction. They form the *action-reaction* force (Newton's 3rd law of motion).
- They are *internal forces*, which are "hidden" in the problem.

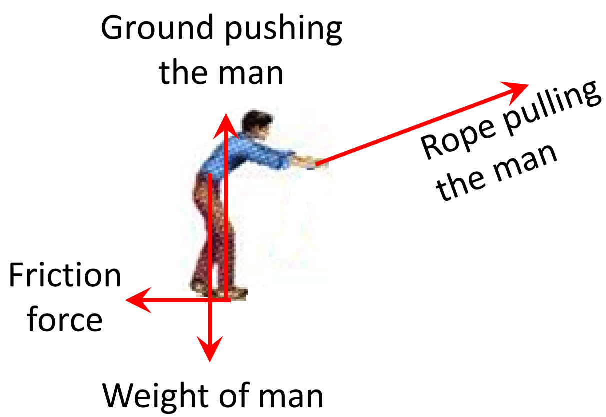
- *Internal forces* - forces that are only holding the particles/bodies forming the rigid body/system together. They have *zero resultant effect* on the body/system because they are action-reaction force pairs.

# Free Body Diagram...

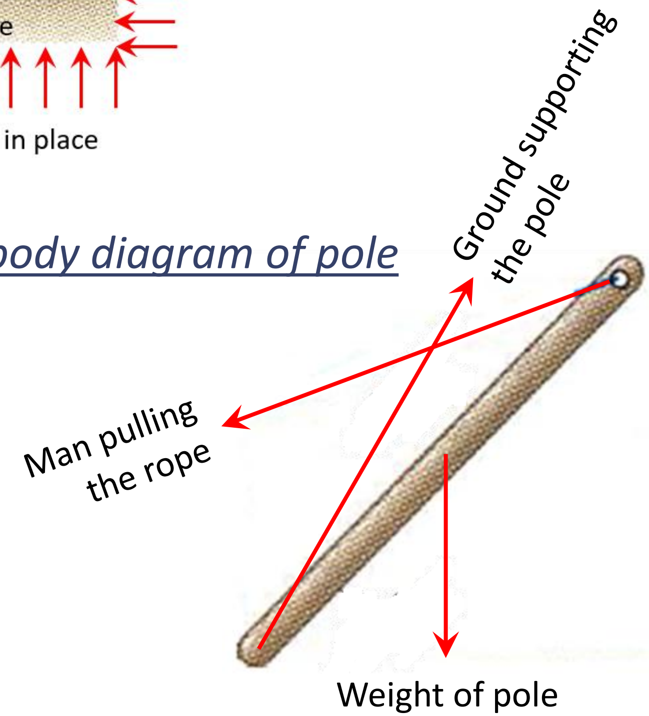
*Free body diagram of entire problem*




*Free body diagram of man*

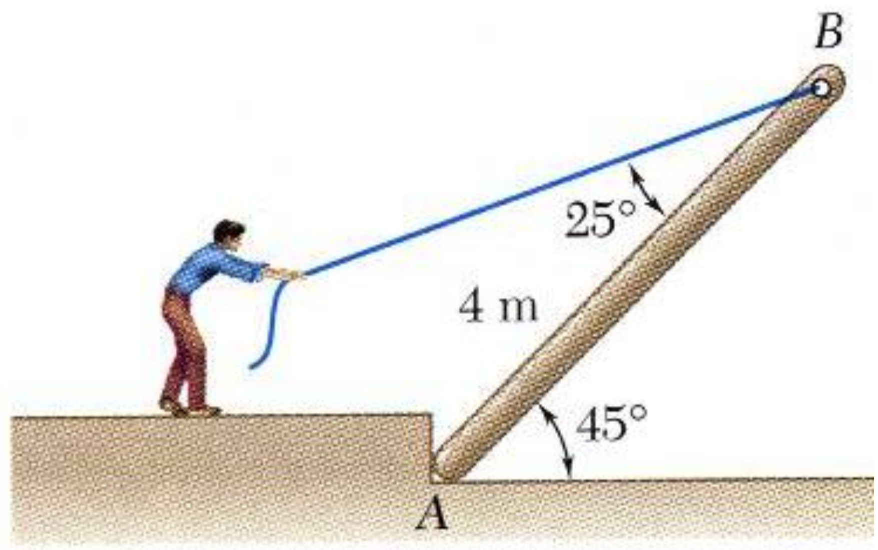



*Free body diagram of pole*



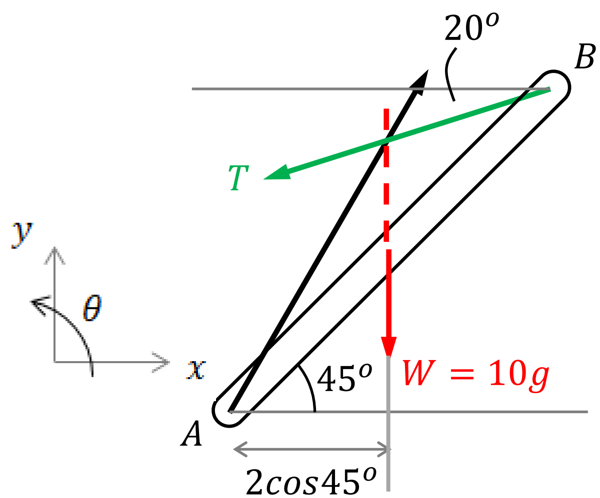
# Example 2.4

A man raises a 10 *kg* pole, of length 4 *m*, by pulling on a rope, and he is holding it stationary at the given position shown. Find the tension in the rope and the magnitude of reaction force at *A.*



# Example 2.4...

*FBD* of the pole




From the geometry,

$$\vec{r}_{B/A} = 4cos45^o\vec{\imath} + 4sin45^o\vec{\jmath}$$

And the tension in the cable is

$$\vec{T} = -Tcos20^o\vec{\imath} - Tsin20^o\vec{\jmath}$$

Taking moment about *A*, we have

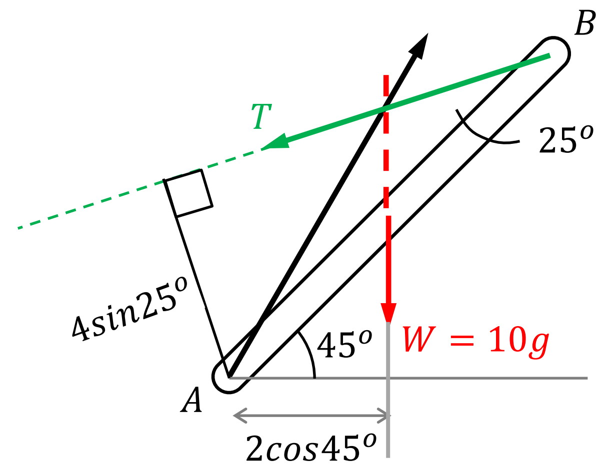
$$\vec{r}_{B/A} \times \vec{T} - (2cos45^o)\,(W) = 0$$

$$\Rightarrow \big((4cos45^o)(-Tsin20^o) - (4sin45^o)(-Tcos20^o)\big) + (2cos45^o)(-10g) = 0$$

$$\Rightarrow T = \mathbf{8.36g}$$

## Example 2.4...

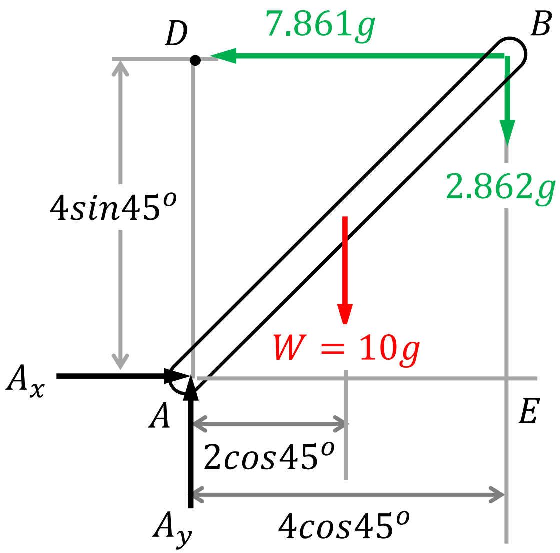
Alternatively, we can identify the perpendicular distant between $\vec{T}$ and point *A*, which is $4sin25^o$.




In this case, the moment equation about *A* becomes

$$T(4sin25^o) - (10g)(2cos45^o) = 0$$

$$\Rightarrow T = \mathbf{8.36g}$$




Updating the tension in the FBD, and from the force equations, we have

$$A_x - Tcos20^o = 0 \Rightarrow A_x = 7.861g$$

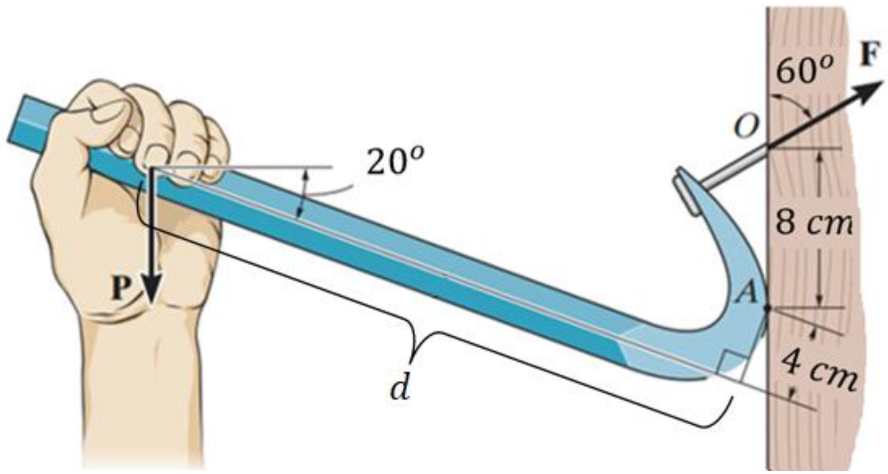
$$A_y - 10g - Tsin20^o = 0 \Rightarrow A_y = 12.862g$$

whose magnitude is given by

$$A = \sqrt{(\mathbf{7.861g})^2 + (\mathbf{12.862g})^2} = \mathbf{15.07g}$$

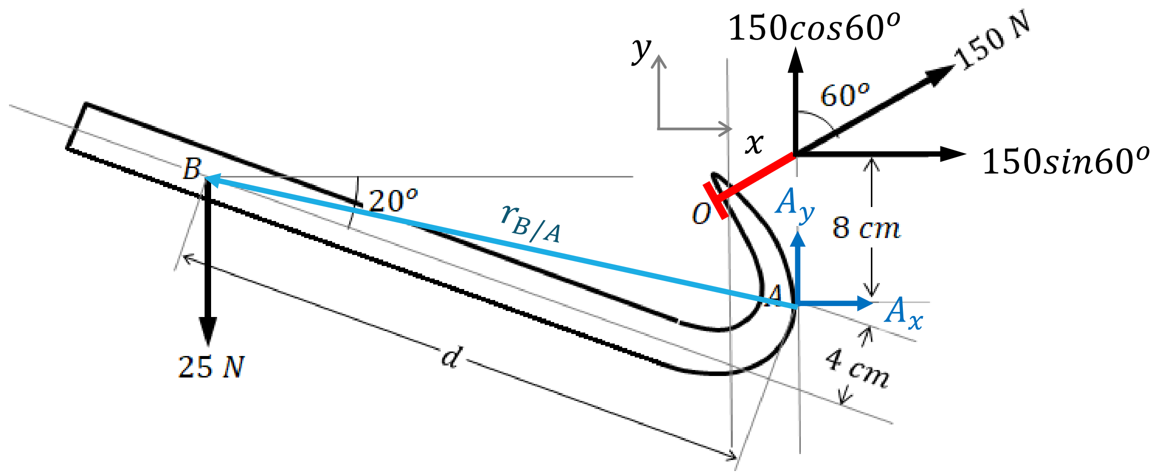
# Example 2.5

The crowbar is subjected to a vertical force of $P$ = 25 $N$ at the grip, and it takes a force of $F$ = 150 $N$ at the claw to pull the nail out. The crowbar contacts the board at point $A$ as shown. Determine where should $P$ (distant $d$ from $A$) be applied to pull out the nail.



# Example 2.5…

Consider the FBD of the crowbar.



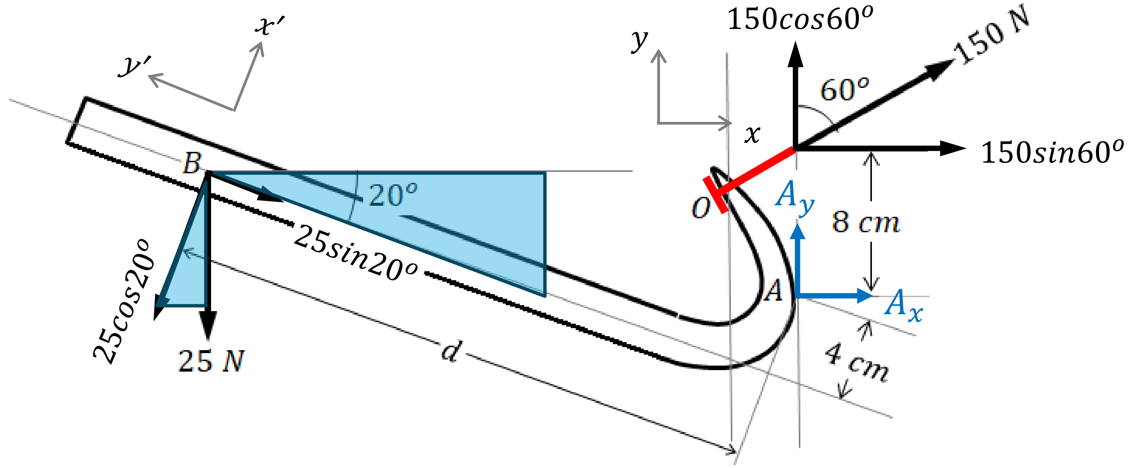

For static equilibrium, one can only use the moment equation about $A$, so that the unknown reaction contact force at $A$ is not involved.

To determine the moment of $P$ about $A$, we need to determine the moment arm vector $r_{B/A}$.

# Example 2.5…

$$\vec{r}_{B/A} = -4\vec{\imath}' + d\vec{\jmath}'$$

$$\vec{F}_{25} = -25cos20^o\vec{\imath}' - 25sin20^o\vec{\jmath}'$$




Note that we can adopt two different coordinate systems to compute the moments the two forces. This is only applicable in *2D planar* problems because the moments are always perpendicular to the 2D plan, as according to the *right-hand screw rule*.

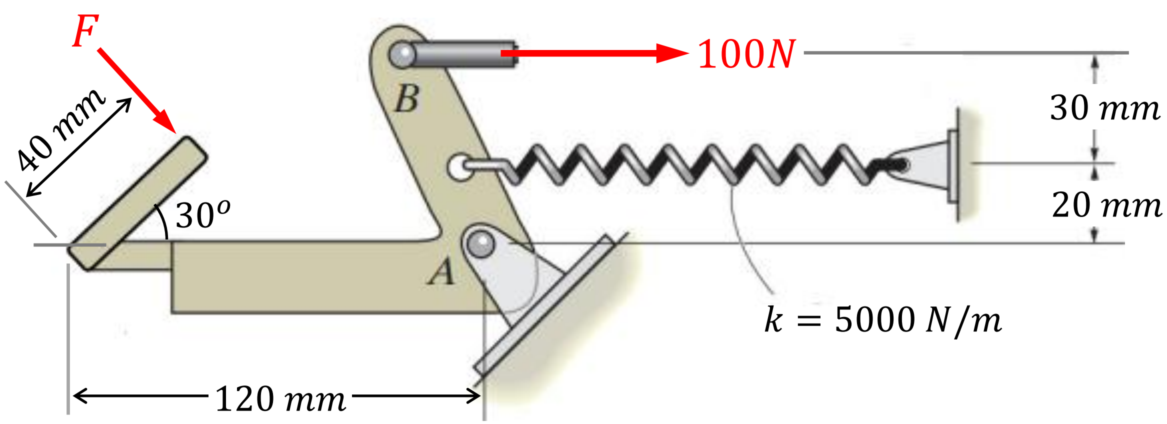
Resolving into the preferred coordinate systems, the moment about *A* gives

$$-(150sin60^o)(8) + \left(-4(-25sin20^o) - d(-25cos20^o)\right) > 0$$

$$\Rightarrow d > \mathbf{42.8\ cm}$$

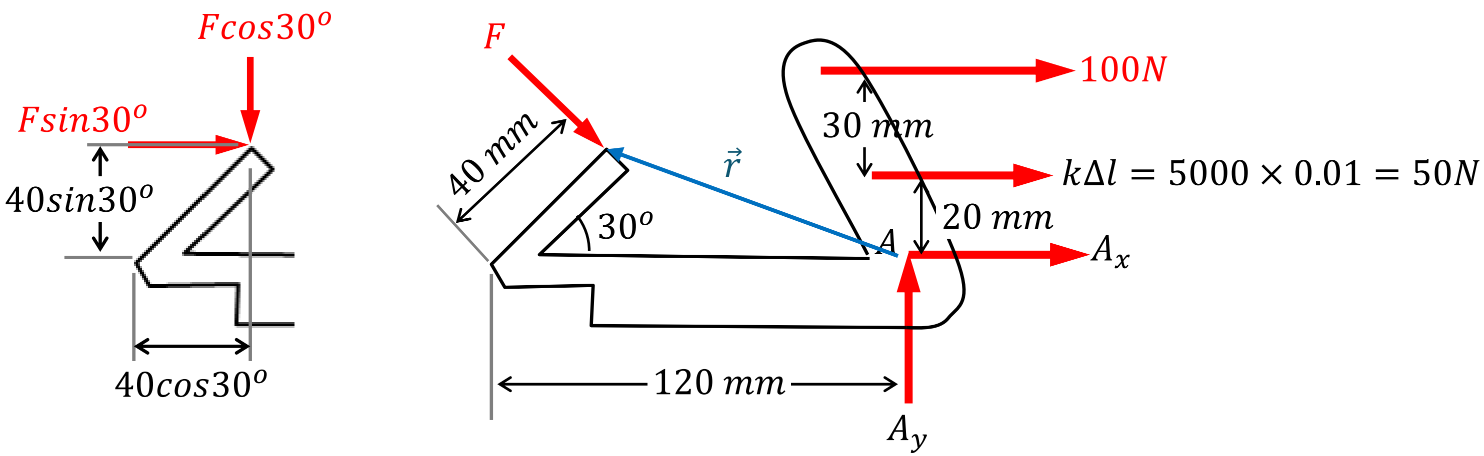
# Example 2.6

An operator applies a vertical force of $F$ on the pedal of the foot lever mechanism as shown. Suppose it is required that the horizontal force generated at $B$ to be 100 $N$ at this configuration, at which the spring is known to be stretched by 1 $cm$, determine the pushing force $F$, and the reaction force at pin $A$.



# Example 2.6...

Consider the FBD of the pedal.




From the geometry, the moment arm vector of the force *F* from *A* is

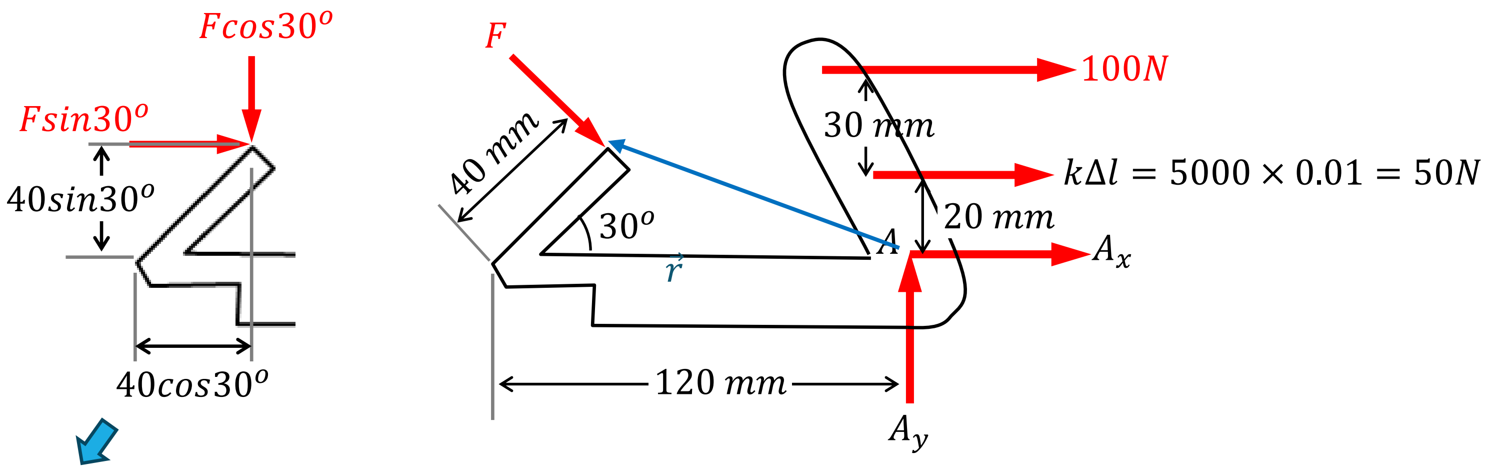
$$\vec{r} = -(0.12 - 0.4cos30^o)\vec{\imath} + 0.4sin30^o\vec{\jmath}$$

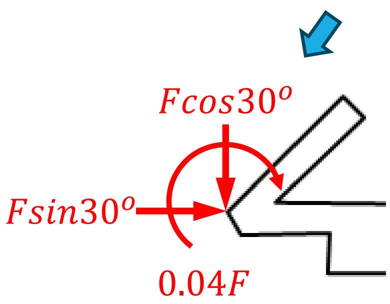
And taking moment about *A*, we have

$$(r_x F_y - r_y F_x) - 100(0.05) - 50(0.02) = 0$$

$$\Rightarrow F = \mathbf{93.86\ N}$$

# Example 2.6…




Taking moment about $A$, we have

$$Fcos30^o(0.12) - 0.04F - 100(0.05) - 50(0.02) = 0$$

$$\Rightarrow F = \mathbf{93.86\ N}$$

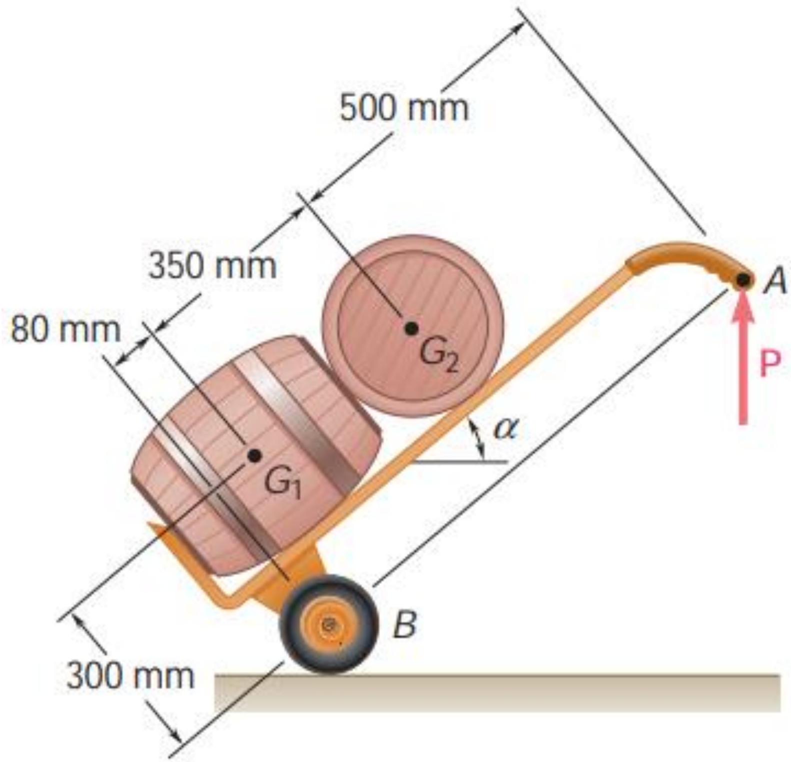
And finally, the force equilibrium in the horizontal and vertical directions, give

$$Fsin30^o + 100 + 50 + A_x = 0 \Rightarrow A_x = \mathbf{-196.93\ N}$$

$$-Fcos30^o + A_y = 0 \Rightarrow A_y = \mathbf{81.29\ N}$$

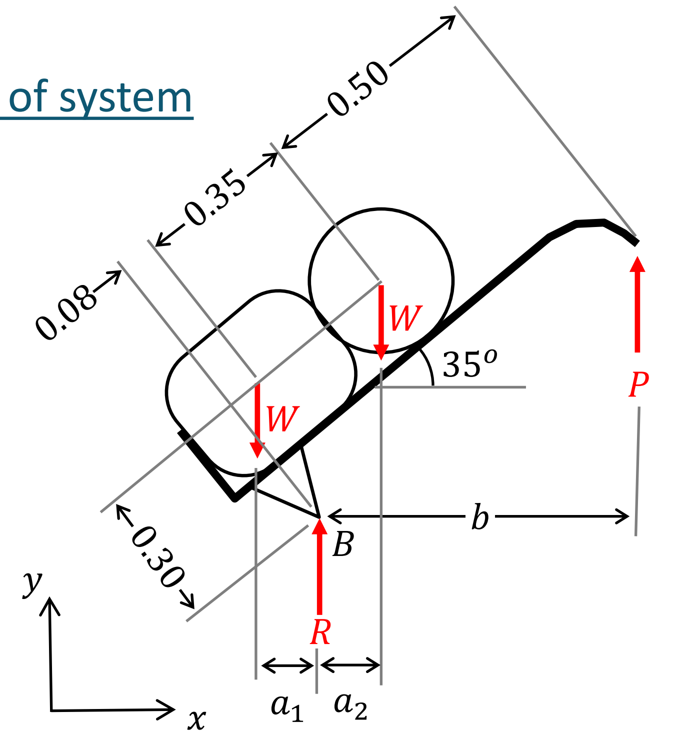
# Example 2.7

A hand truck is used to move two barrels, each of mass 40 *kg*. Neglecting the mass of the hand truck, determine the vertical force *P* that should be applied to the handle to maintain equilibrium when $\alpha$ = 35°.



# Example 2.7…

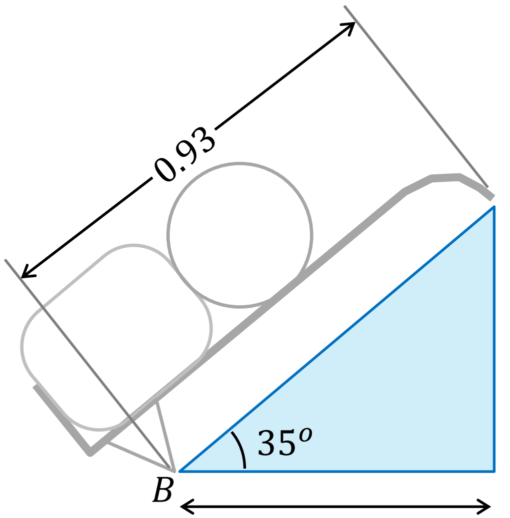
FBD of system



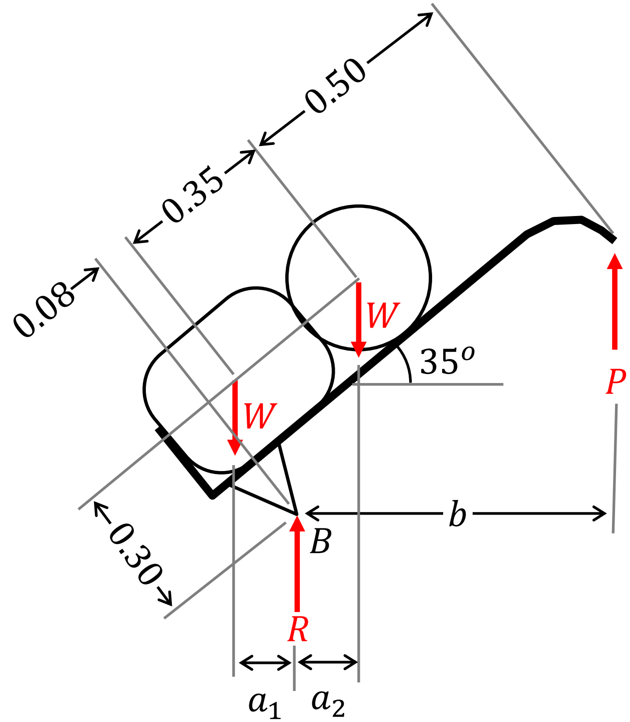

Consider the moment about *B*, we have

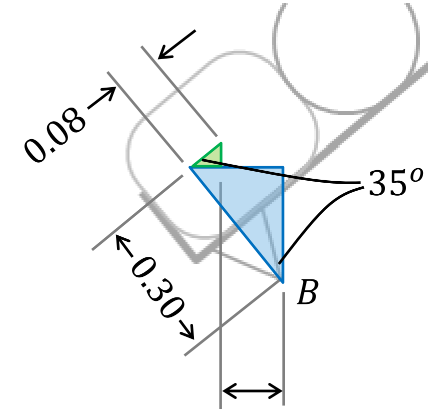
$$Pb - Wa_2 + Wa_1 = 0 \Rightarrow P = \frac{W}{b}(a_2 - a_1)$$

# Example 2.7…

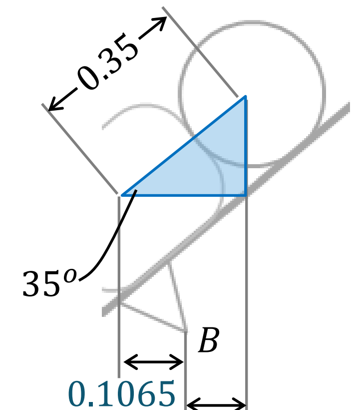



$0.93cos35^o = \mathbf{0.7618}$




$0.30sin35^o - 0.08cos35^o = \mathbf{0.1065}$

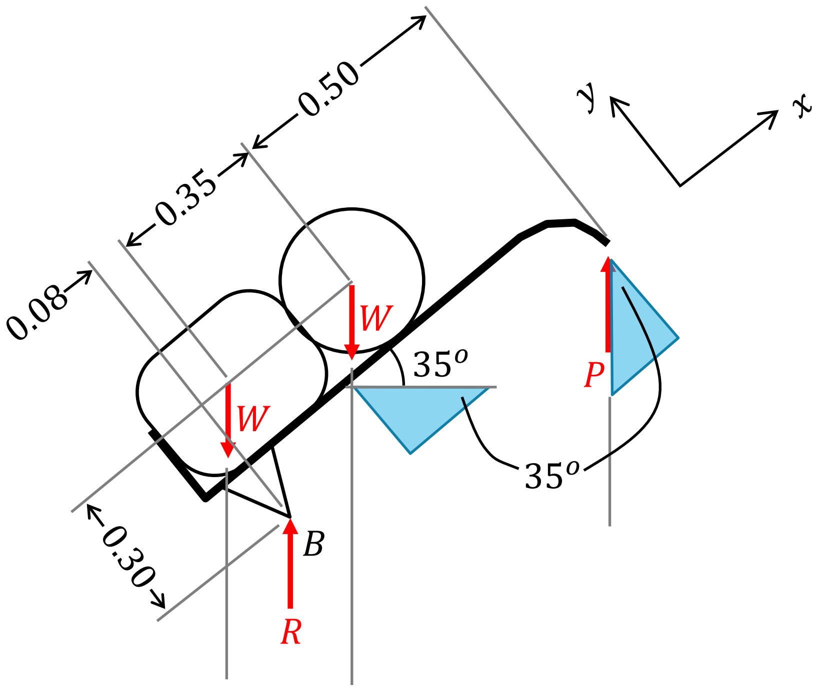



$0.35cos35^o - 0.1065 = \mathbf{0.1802}$

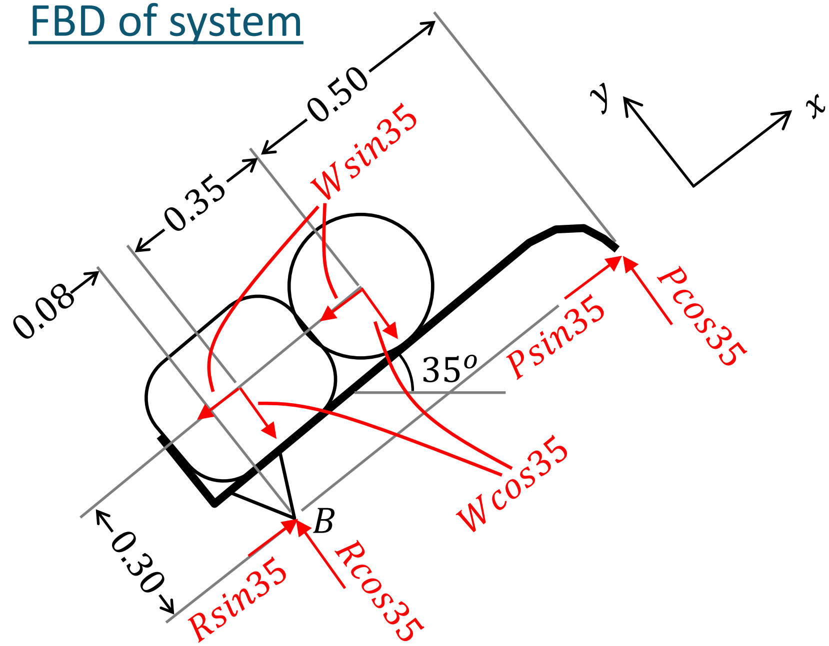
Finally, substituting the numeric into the moment equation, gives

$$P = \frac{40g}{0.7618}(0.1802 - 0.1065) = \mathbf{37.96\ N}$$

# Example 2.7…




FBD of system




Taking moment about $B$, gives

$$Pcos35(0.93) + 2Wsin35(0.3) - Wcos35(0.08) - Wcos35(0.08 + 0.35) = 0$$

$$\Rightarrow P = \mathbf{37.92\ N}$$

# Reactions at Supports and Connections in 2D

- Rigid body needs to be supported at the connecting points to stay in equilibrium state. They are *internal forces* before isolating the body.

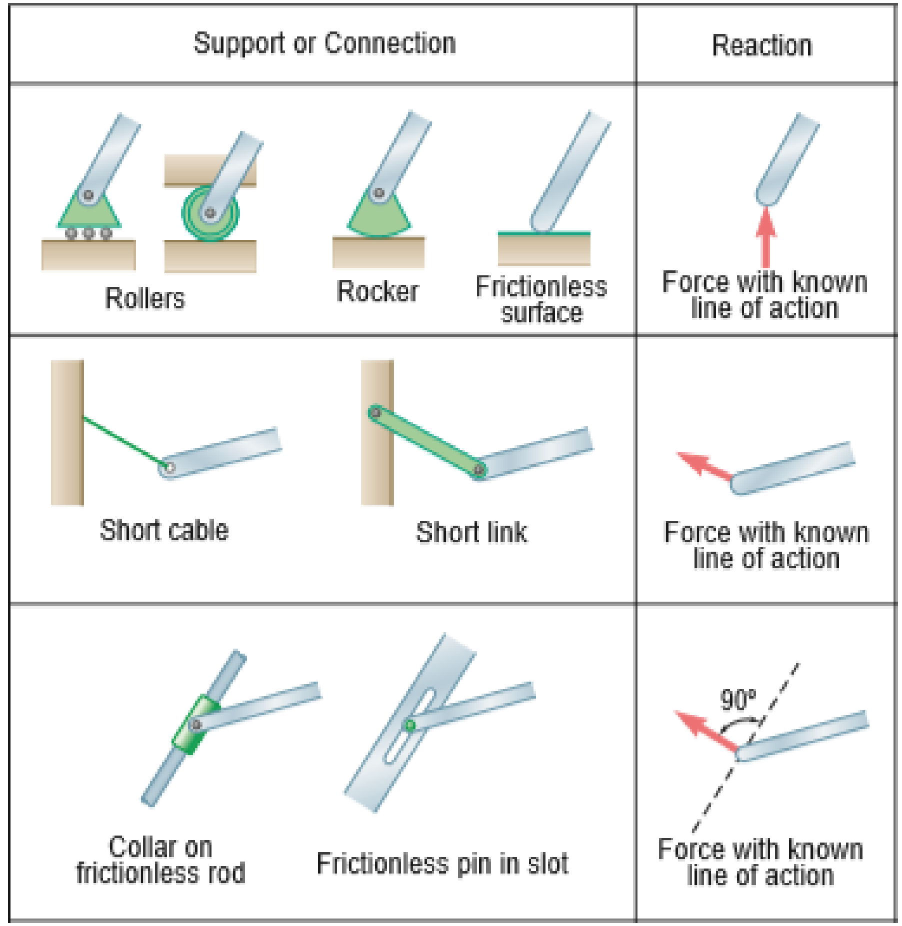
- When the rigid body is isolated in *FBD*, these supports must be replaced by its equivalent reaction forces/moment.

- There are three types of supports or connections:

• ***Reaction force with known line of action****:*

These supports and connections constraint motion in one direction only. In other words, the *direction* of the reaction force is *known* but *not* its *magnitude*.

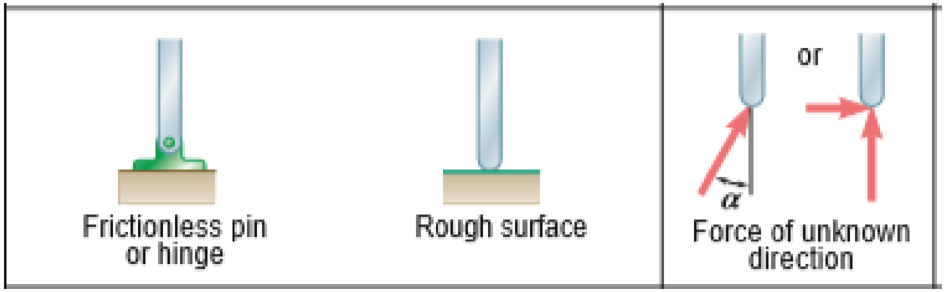
Hence, this group involves only *one unknown*.



# Reactions at Supports and Connections in 2D…

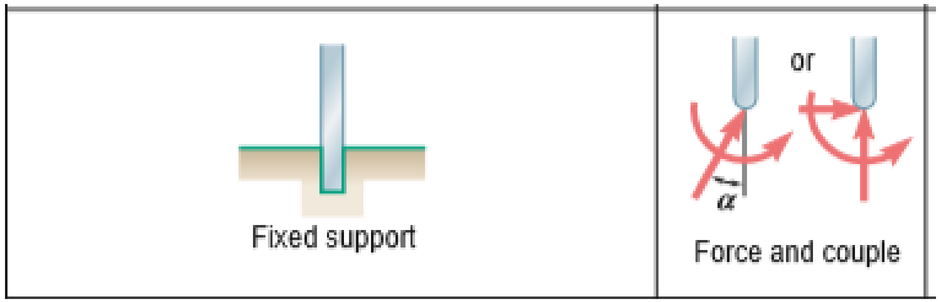
- ***Reaction force with unknown direction and magnitude:***

  They <u>*prevent the translation*</u> of the rigid body in all directions but allow rotation in the connection. Hence, reaction for this group involves <u>*two unknowns*</u>, usually in the *x*- and *y*- directions in the rectangular coordinates.




- ***Reaction is a force-couple pair:***

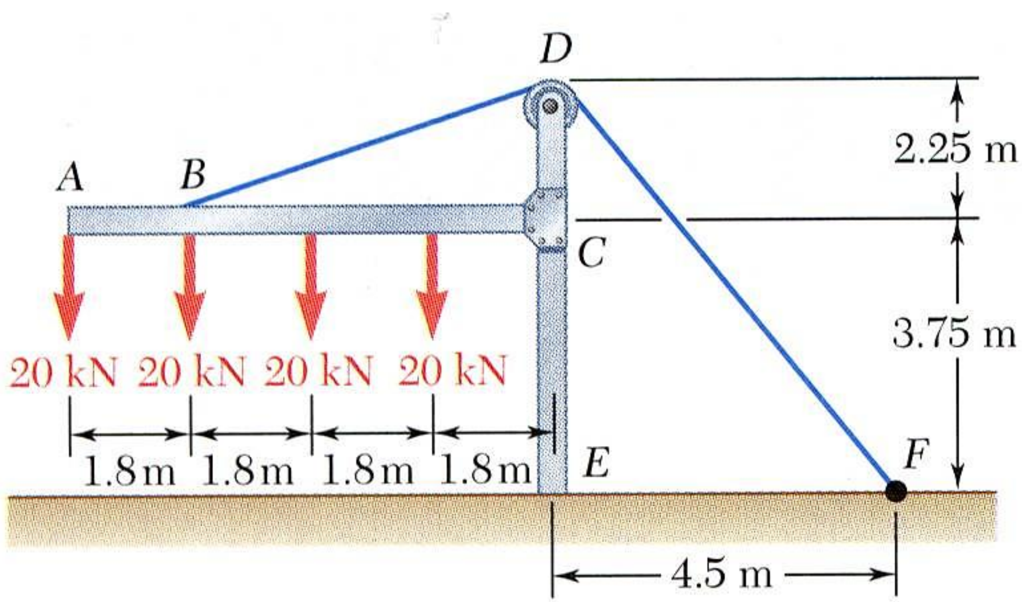
  These reactions are caused by <u>*fixed*</u> supports, which constrained the body from moving completely. Hence, the reactions for this group involve <u>*three unknowns*</u>, namely the forces in the two directions, and a couple.



# Example 2.8

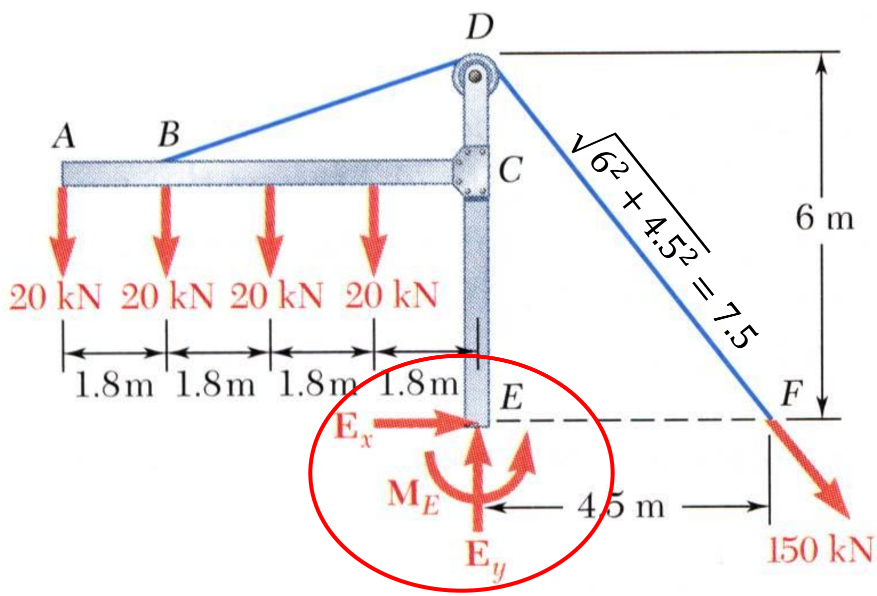
The frame supports part of the roof of a small building. The tension in the cable is 150 *kN*.

Determine the reaction at the fixed end *E*.



# Example 2.8…

Free-body diagram




Due to the fixed constraint, the reaction at *E* comprises three unknowns.

The tension in the cable can be decomposed into:

$$T_x = 150\left(\frac{4.5}{7.5}\right) = 90\ kN, and$$

$$T_y = 150\left(\frac{6}{7.5}\right) = 120\ kN$$

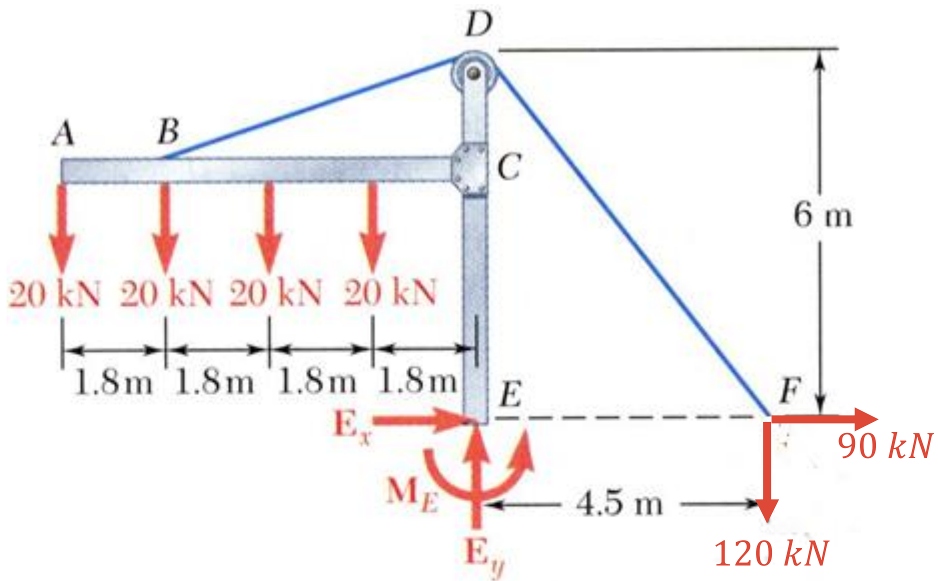
Consider the force equilibrium equations, we have

$$E_x + T_x = 0 \Rightarrow E_x = \mathbf{-90\ kN}$$

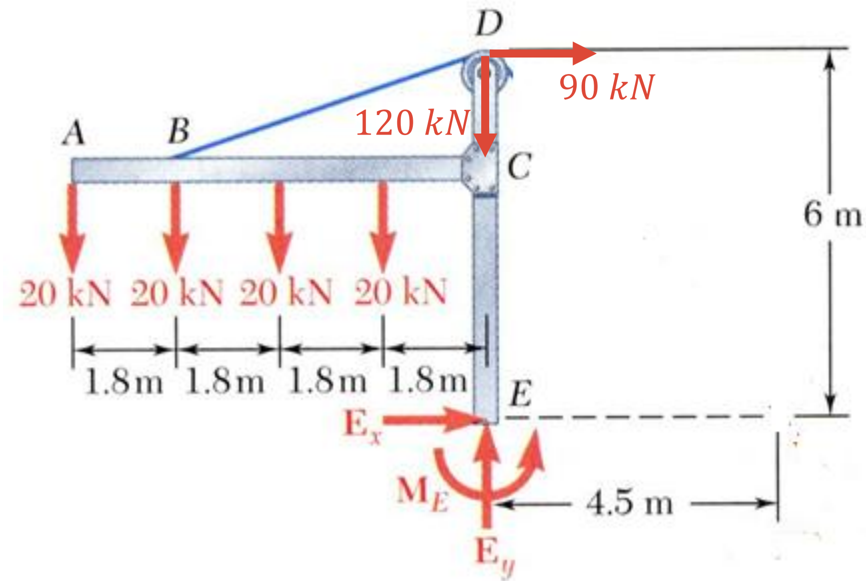
$$E_y - 4(20) - T_y = 0 \Rightarrow E_y = \mathbf{200\ kN}$$

# Example 2.8…

Free-body diagram




Free-body diagram




The moment about *E* is

$$M_E + (20)(7.2) + (20)(5.4) + (20)(3.6) + (20)(1.8) - T_y(4.5) = 0;$$

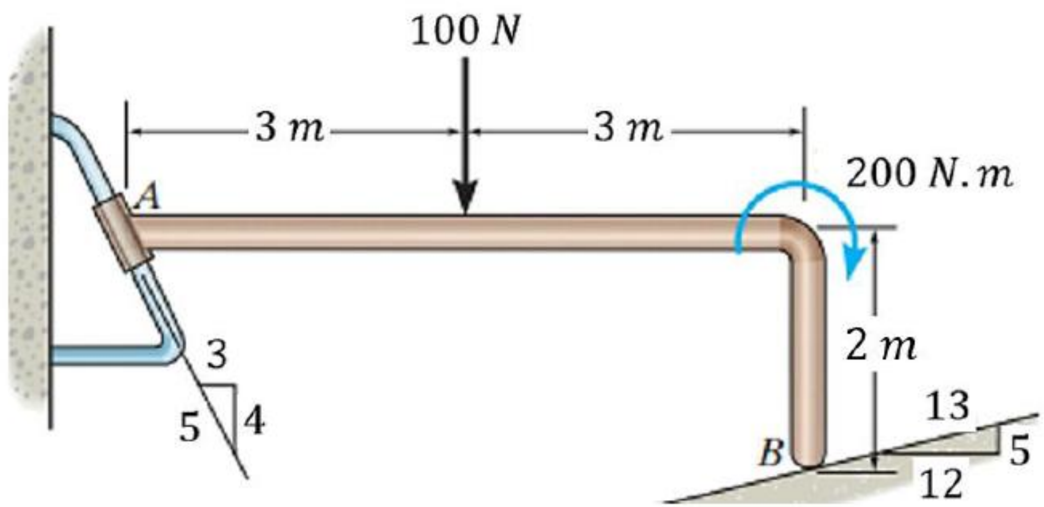
$or$

$$M_E + (20)(7.2) + (20)(5.4) + (20)(3.6) + (20)(1.8) - T_x(6.0) = 0$$

$$\Rightarrow M_E = \mathbf{180\ kNm}$$

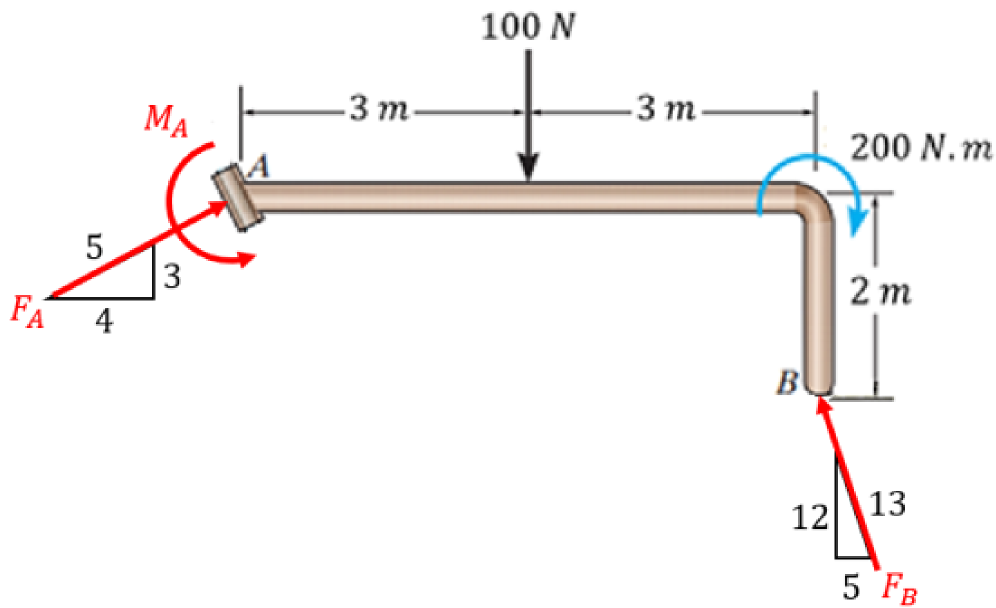
# Example 2.9

The bent rod *AB* is supported by a smooth surface at *B* and by a collar at *A*, at which it is free to slide along the fixed guiding rod. Determine the constraints imposed on rod *AB* by the guiding rod at *A*.



# Example 2.9...

Consider the FBD of rod *AB*.

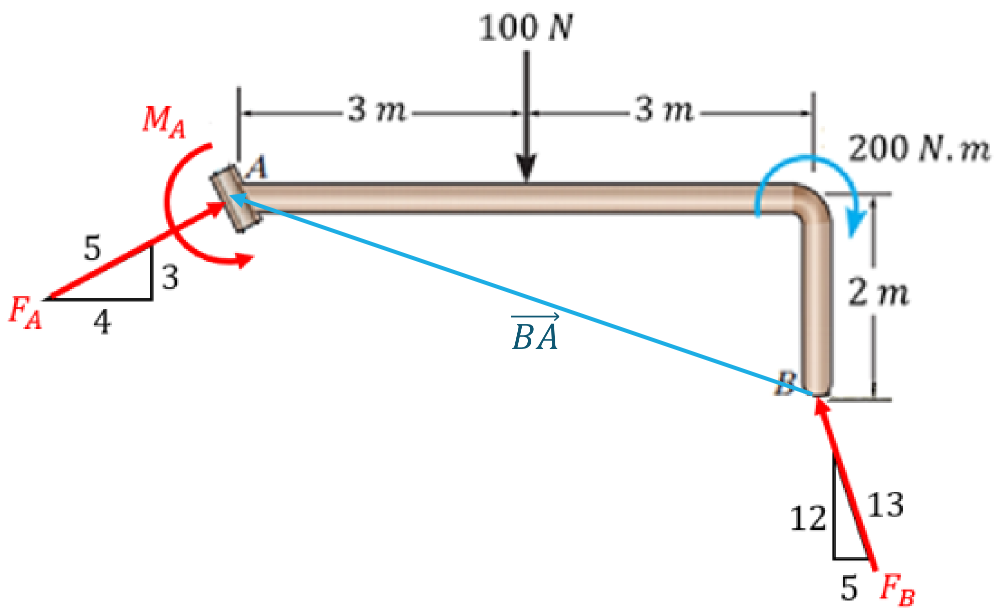



Using the force equations, we have

$$\sum F_x = 0\,, \qquad F_A\left(\frac{4}{5}\right) - F_B\left(\frac{5}{13}\right) = 0 \Rightarrow F_B = \left(\frac{52}{25}\right) F_A$$

$$\sum F_y = 0\,, \qquad F_A\left(\frac{3}{5}\right) + F_B\left(\frac{12}{13}\right) - 100 = 0 \Rightarrow F_A = 39.68\ N$$

# Example 2.9...




Force $F_A$ is

$$\overrightarrow{F_A} = \left(\frac{4}{5}F_A\right)\vec{\imath} + \left(\frac{3}{5}F_A\right)\vec{\jmath}$$

$$= 31.74\vec{\imath} + 23.81\vec{\jmath}$$

Moment arm vector of $F_A$ about B is

$$\overrightarrow{BA} = -6\vec{\imath} + 2\vec{\jmath}$$

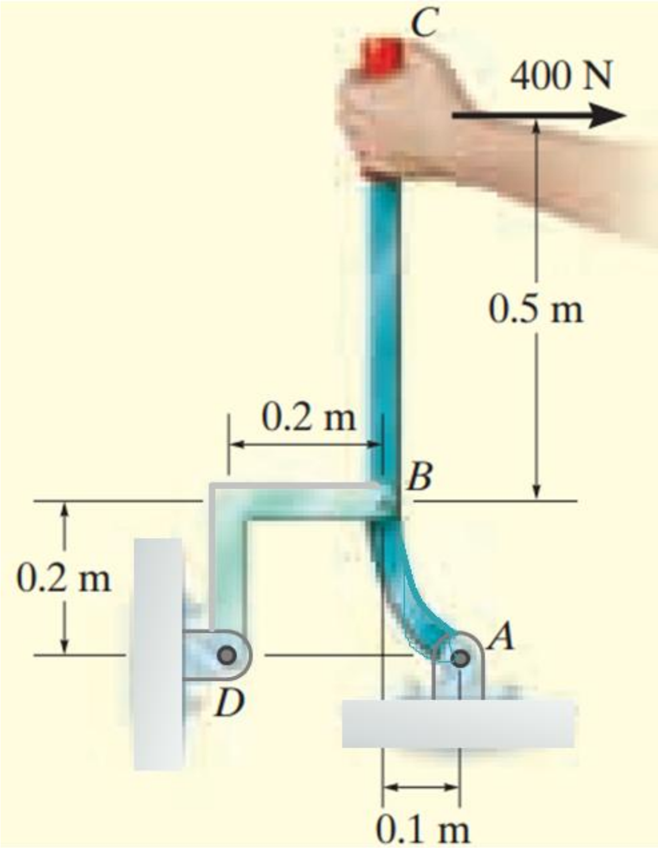
Next, taking moment about $B$, gives

$$\sum M_B = 0\,, \qquad M_A + 100(3) - 200 + \big((-6)(23.81) - (2)(31.74)\big) = 0$$

$$\Rightarrow M_A = \mathbf{106\ N.m}$$

# Example 2.10

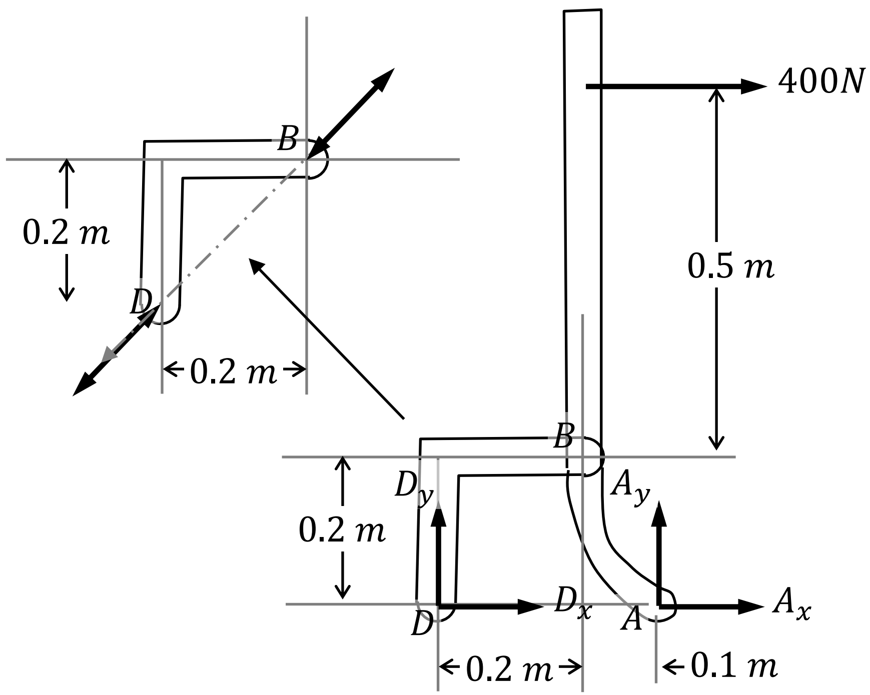
The lever *ABC* is pin supported at *A* and connected to a short link *BD* as shown in the figure. Assume the weight of the members are negligible, determine the reaction force of the pin on the lever at *A*.

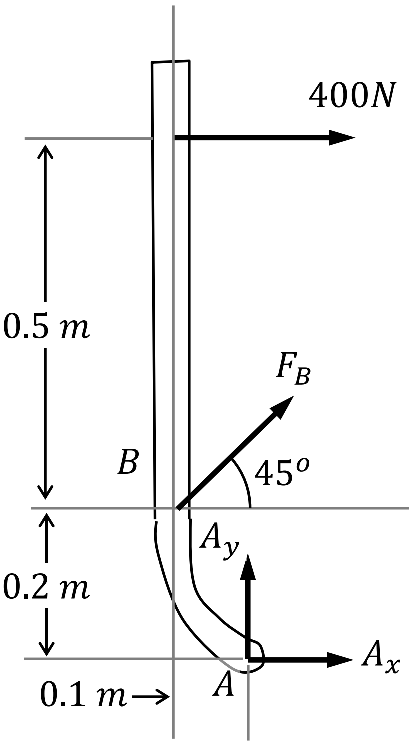


# Example 2.10...

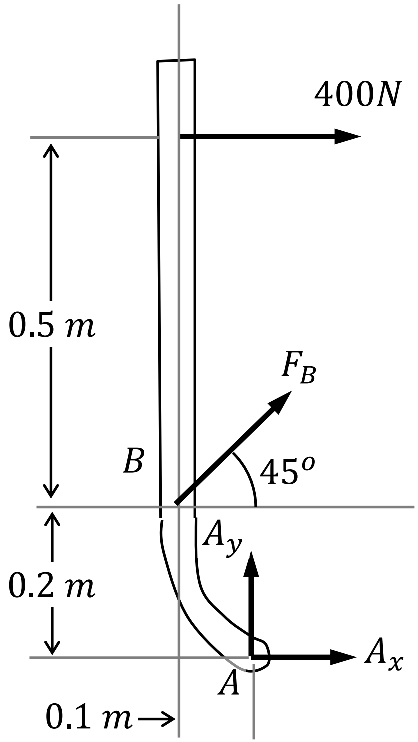
Consider the *FBD* of the entire assembly. It shows <u>*4 unknowns*</u>. Hence, we cannot solve this problem yet because we can only have 3 <u>*independent equations*</u> from static analysis...

Unless we recognise that member *BD* is a 2-force body.



# Example 2.10…




Taking moment about $B$, gives

$$-400(0.5) + A_x(0.2) + A_y(0.1) = 0$$

$$\Rightarrow 2A_x + A_y = 2000$$

And consider force equilibrium in *x*- and y- directions,

$$A_y + F_B sin45^o = 0 \Rightarrow A_y = -F_B sin45^o$$

$$400 + A_x + F_B cos45^o = 0 \Rightarrow -F_B cos45^o - A_x = 400$$

$$\Rightarrow A_y - A_x = 400$$

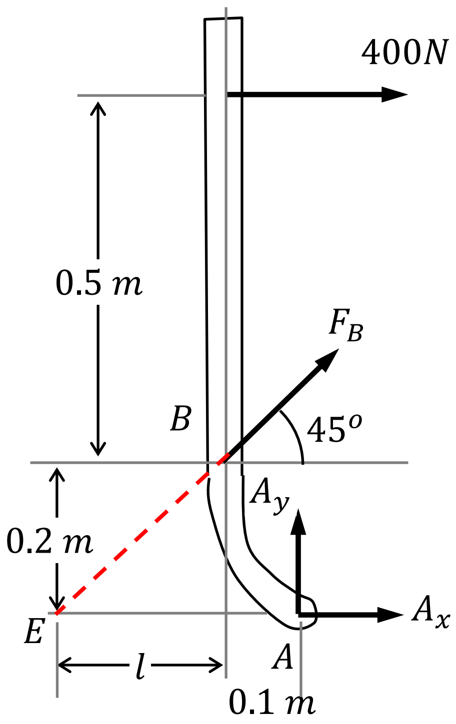
Solving the two equations simultaneously, we have

$$A_x = \mathbf{533.3\ N} \quad \text{and} \quad A_y = \mathbf{933.3\ N}$$

$$\Rightarrow\ F_A = \sqrt{533.3^2 + 933.3^2} = \mathbf{1075\ N}$$

# Example 2.10…

Alternatively, by projecting along the line-of-actions of $F_B$ and $A_x$, we get the interception point $E$.




From geometry,

$$tan45^o = \frac{0.2}{l} \Rightarrow l = 0.2$$

With this, taking moment about $E$, we have

$$A_y(0.2 + 0.1) - 400(0.7) = 0$$

$$\Rightarrow A_y = \mathbf{933.3\ N}$$

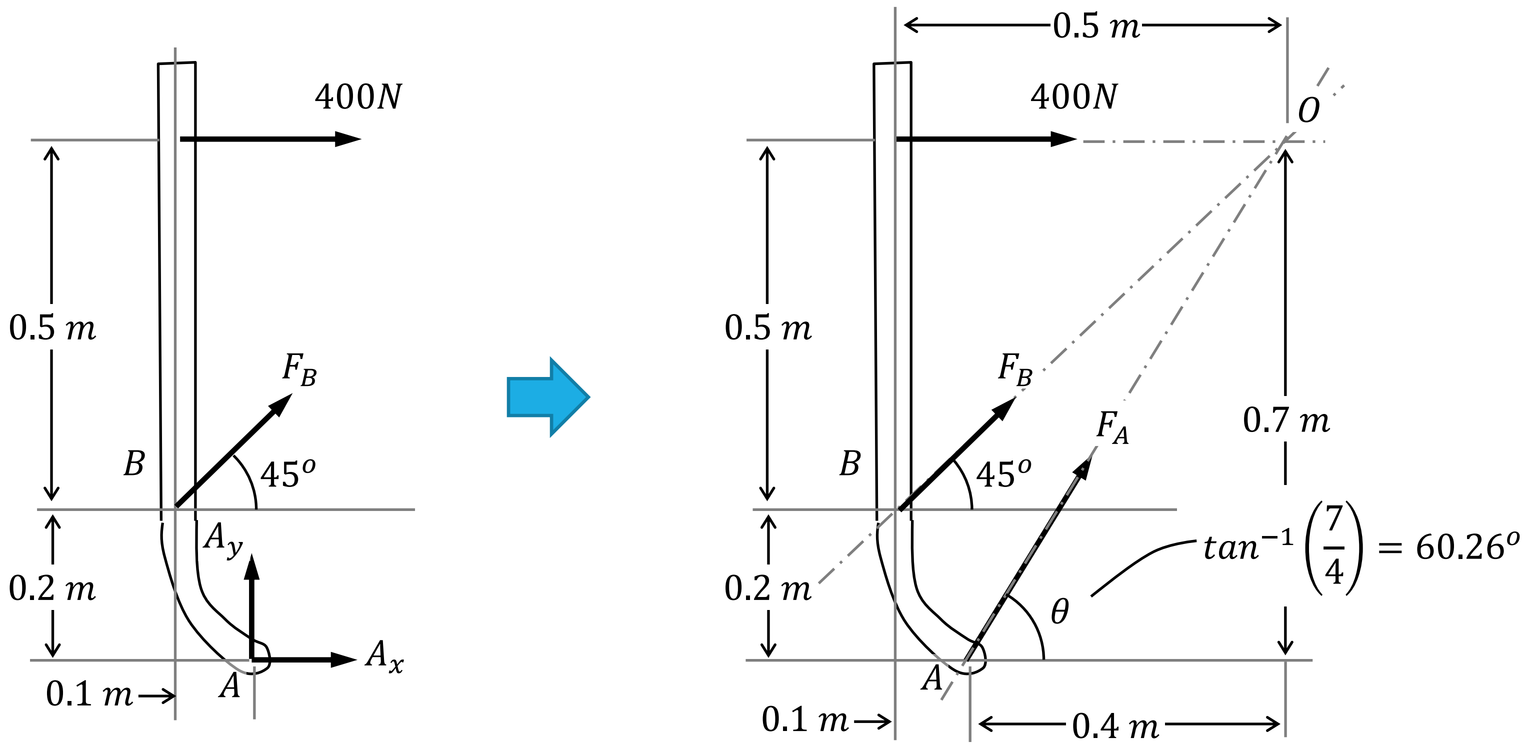
And then followed by taking moment about $B$, gets

$$933.3(0.1) + A_x(0.2) - 400(0.5) = 0$$

$$\Rightarrow A_x = \mathbf{533.3\ N}$$

# Example 2.10…

Another approach is based on the 3-force body concept.

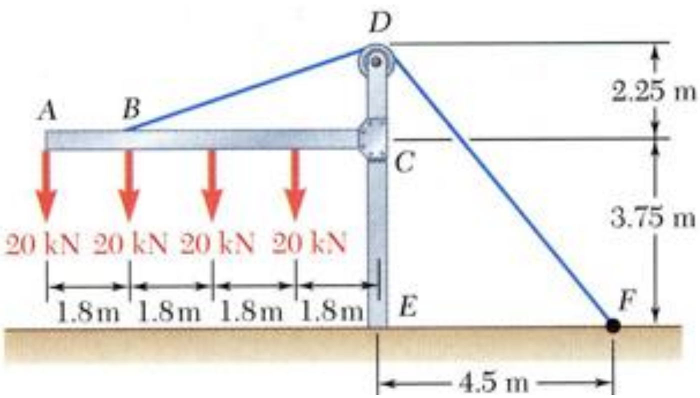


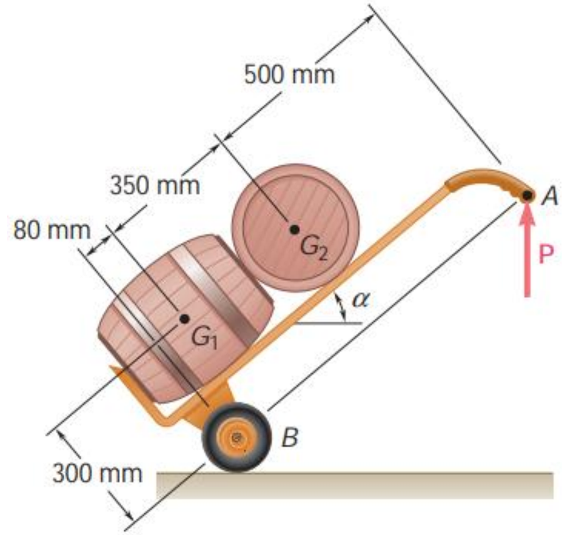

Taking moment about *B*, gives

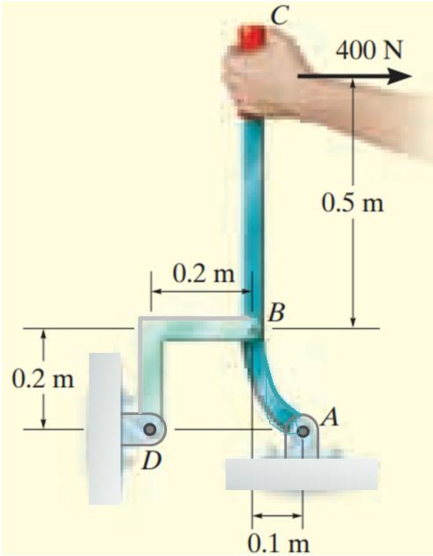
$$-400(0.5) + (F_A cos60.26^o)(0.2) + (F_A sin60.26^o)(0.1) = 0 \Rightarrow F_A = \mathbf{1075\ N}$$

# Constraints and Statical Determinacy

- So far, the supports in examples were such that the body/assembly could not move under the given loads. They are said to be *completely constrained* and hence the problem is *well-posed*.



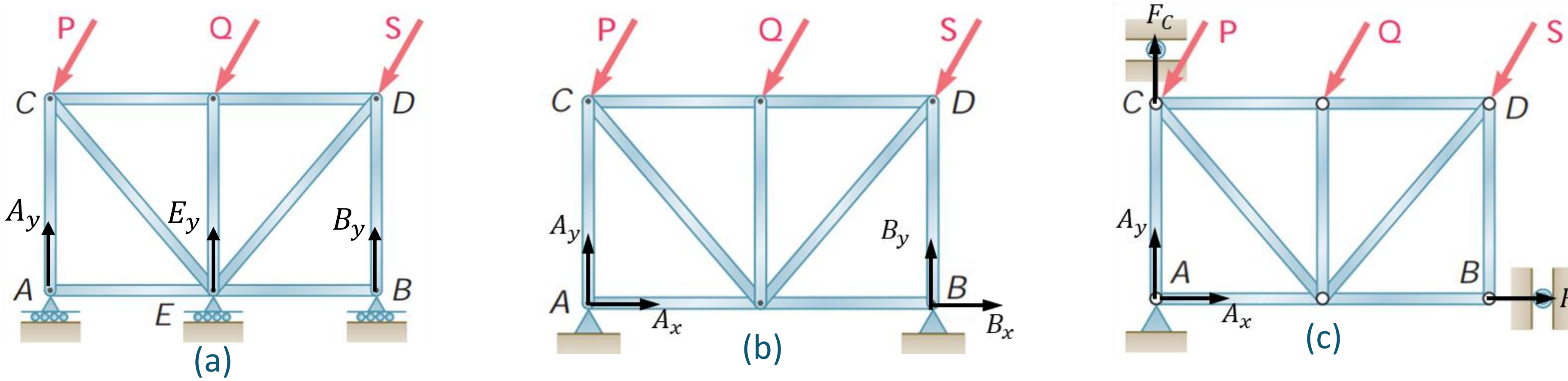




- For these examples, it an be noted that the *number of unknowns is equal to the number of equations*. They can be determined by *static analysis alone*, and hence they are referred to as *statically determinate* problems.

# Constraints and Statical Determinacy…

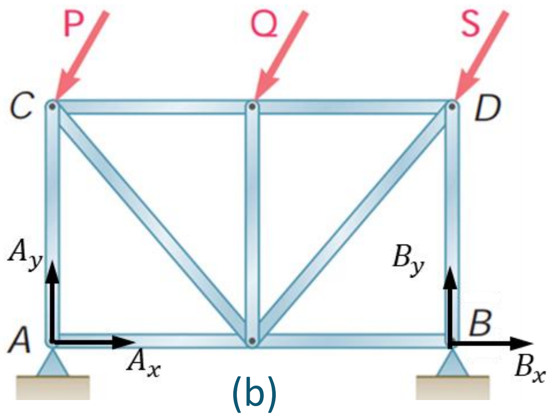
- Let us consider a truss that is supported differently as follows.




- Based on the number of unknowns, one may conclude that:
  - Case (a) is completely-constrained, and hence it can be solved by statics;
  - Cases (b) and (c) are *over-constrained* since it involves more than 3 unknowns, and hence it cannot to solve statically.

- However, cases (a) and (c) turn out to be *geometric instability* configurations. That is, the truss will move left in (a), whereas in (c), it will rotate freely about *A* if the effective moment of *P*, *Q* and *R* is not zero.

# Constraints and Statical Determinacy…




(b)

- On the other hand, (b) is over-constrained, and hence is *statically indeterminate.*
- This problem can only be resolved by finding another equation (such as imposing *compatibility conditions*) to define the problem completely, i.e.

*No. of independent equations = No. of unknowns*

- In summary, the above this condition, while *necessary*, is *not sufficient* to ensure that the reactions are *statically determinate*, such as in case (a).